

Chapter 8

Quadrature Amplitude Modulation

At this point, all the passband modulation schemes we have studied, FSK, PSK, CPM, and MHPM, are constant envelope schemes. The constant envelope property of these schemes is specially important to systems with power amplifiers which must operate in the nonlinear region of the input-output characteristic for maximum power efficiency, like the satellite transponders. For some other communication systems, constant envelope may not be a crucial requirement, whereas bandwidth efficiency is more important. Quadrature amplitude modulation (QAM) is such a class of non-constant envelope schemes that can achieve higher bandwidth efficiency than MPSK with the same average signal power. QAM is widely used in modems designed for telephone channels. The CCITT telephone circuit modem standards V.29 to V.33 are all based on various QAM schemes ranging from uncoded 16-QAM to trellis coded 128-QAM. The research of QAM applications in satellite systems, point-to-point wireless systems, and mobile cellular telephone systems also has been very active.

In Chapter 1 we mentioned binary amplitude shift keying (ASK). ASK can also be made M-ary, called M-ary amplitude modulation (MAM). MAM is usually no longer a preferable choice due to its poor power efficiency. However, since QAM signal consists of two MAM components and they can be demodulated in two separate channels, it is necessary to understand MAM's behavior in order to understand QAM. We discuss MAM in Section 8.1. Then we move on to define QAM signal and constellation in Section 8.2. Various QAM constellations are introduced in Section 8.3, but only the square QAM constellations are described in detail. QAM's PSD, modulator, demodulator, error probability, synchronization, and differential coding are discussed in Sections 8.4 to 8.9. Section 8.10 summarizes the chapter.

8.1 M-ARY AMPLITUDE MODULATION

In this section we first introduce MAM in its most general form which is applicable to

baseband as well as bandpass signals. Then we focus our attention on the analysis of bandpass MAM signals which will serve as a foundation for analyzing QAM signals.

In its most general form, an M-ary amplitude modulation signal can be expressed as

$$s_i(t) = s_i\phi(t), \quad 0 \leq t \leq T \quad (8.1)$$

for $i = 1, 2, \dots, M$, $\phi(t)$ is any unit energy function of duration T , that is $\int_0^T \phi^2(t)dt = 1$. Consequently $s_i^2 = E_i$ is the energy of $s_i(t)$. If $\phi(t)$ is a baseband pulse, then $s_i(t)$ is the baseband M-ary amplitude modulation, which is usually called M-ary pulse amplitude modulation (PAM). The NRZ and RZ line codes are examples of baseband binary AM schemes. If $\phi(t)$ is a high frequency sinusoidal carrier, then $s_i(t)$ is the passband M-ary amplitude modulation. The passband MAM is also called amplitude shift keying (ASK). The OOK (on-off keying) is a binary passband AM scheme with one of the s_i being zero. The BPSK can be also viewed as a binary passband AM scheme with two antipodal s_i .

8.1.1 Power Spectral Density

On the entire time axis, we can write MAM signal as

$$s(t) = \sum_{k=-\infty}^{\infty} s_k\phi(t - kT), \quad -\infty < t < \infty \quad (8.2)$$

where amplitude s_k is determined by the message data which are random. Thus s_k is a random variable. Expression (8.2) is in the form of (A.14) in Appendix A. Further assuming that data are uncorrelated, the PSD of $s(t)$ is given by (A.18), that is

$$\Psi_s(f) = \frac{\sigma_s^2 |\Phi(f)|^2}{T} + \left(\frac{m_s}{T}\right)^2 \sum_{k=-\infty}^{\infty} |\Phi(\frac{k}{T})|^2 \delta(f - \frac{k}{T}) \quad (8.3)$$

where $\Phi(f) = \mathcal{F}\{\phi(t)\}$, σ_s^2 is the variance of s_k and m_s is the mean value of s_k . The first part of (8.3) is the continuous spectrum and the second part is the discrete spectral lines. Note that $\Phi(f)$ could be a baseband spectrum or a bandpass spectrum, depending on whether $\phi(t)$ is baseband or bandpass

For MAM, usually the amplitudes are uniformly spaced and symmetrically located around zero. Then $m_s = 0$, thus

$$\Psi_s(f) = \frac{\sigma_s^2 |\Phi(f)|^2}{T} \quad (8.4)$$

This tells us that the PSD of MAM is determined by the PSD of the basis function

$\phi(t)$. If the amplitude distribution is not symmetrical around zero, then $m_s \neq 0$, discrete spectral lines will be present.

Note that (8.3) and (8.4) are applicable to both baseband and bandpass $\phi(t)$. However, for bandpass $\phi(t)$, the spectrum $\Phi(f)$ has two parts, centered around f_c and $-f_c$. Thus (8.3) and (8.4) are not convenient to use. For bandpass random signal the PSD is completely determined by the PSD of its complex envelope or equivalent baseband signal (A.13). Thus it suffices to have the PSD of the complex envelope $\Psi_{\tilde{s}}(f)$ instead of $\Psi_s(f)$. Assuming the bandpass MAM signal is

$$s_i(t) = A_i p(t) \cos 2\pi f_c t, \quad 0 \leq t \leq T$$

where $p(t)$ is a pulse-shaping function, then the complex envelope of $s(t)$ on the entire time axis is

$$\tilde{s}(t) = \sum_{k=-\infty}^{\infty} A_k p(t - kT), \quad -\infty < t < \infty$$

which is in the form of (A.14). Thus for uncorrelated data, the PSD of bandpass MAM signal is

$$\Psi_{\tilde{s}}(f) = \frac{\sigma_A^2 |P(f)|^2}{T} + \left(\frac{m_A}{T}\right)^2 \sum_{k=-\infty}^{\infty} |P(\frac{k}{T})|^2 \delta(f - \frac{k}{T}) \quad (8.5)$$

where $P(f) = \mathcal{F}\{p(t)\}$ is the spectrum of $p(t)$, σ_A^2 is the variance of A_k , and m_A is the mean value of A_k . Usually the amplitudes are uniformly spaced and symmetrically located around zero. Then $m_A = 0$, and

$$\Psi_{\tilde{s}}(f) = \frac{\sigma_A^2 |P(f)|^2}{T} \quad (8.6)$$

The above shows that the PSD of bandpass MAM is determined by the PSD of the pulse-shaping function $p(t)$. If the amplitude distribution is not symmetrical around zero, then $m_A \neq 0$, discrete spectral lines will be present, such as in the OOK case.

Now let us assume $p(t)$ is rectangular with unit amplitude then

$$|P(f)| = \left| T \frac{\sin \pi f T}{\pi f T} \right| \quad (8.7)$$

and

$$\Psi_{\tilde{s}}(f) = \sigma_A^2 T \left(\frac{\sin \pi f T}{\pi f T} \right)^2 \quad (8.8)$$

which has the same shape of the PSD of MPSK.

8.1.2 Optimum Detection and Error Probability

In the M-ary PAM signal set, there is only one basis function which is the $\phi(t)$. Geometrically, each signal can be represented by its projection on the $\phi(t)$

$$s_i = \int_0^T s_i(t) \phi(t) dt$$

Thus from (B.37), the optimum detector is a one-dimensional minimum distance detector. Assuming the received signal is

$$r(t) = s_i(t) + n(t)$$

the sufficient statistic is

$$r = \int_0^T r(t) \phi(t) dt \quad (8.9)$$

and the detector compares r to s_i and chooses the closest (minimum distance decision rule).

The error probability of the coherent detection for an M-ary PAM with equal amplitude spacings can be derived as follows. Assuming AWGN channel with two-sided noise PSD of $N_o/2$,

$$r = \int_0^T r(t) \phi(t) dt = \int_0^T [s_i(t) + n(t)] \phi(t) dt = s_i + n$$

where n is Gaussian with zero mean and a variance of $N_o/2$. Thus r is Gaussian with mean s_i and variance $N_o/2$. Figure 8.1 shows the probability distribution densities of r conditioned on s_i , where τ_i are thresholds. This figure can help us derive the error probability.

Assuming s_i is transmitted, a symbol error occurs when the noise n exceeds in magnitude one-half of the distance between two adjacent levels. This probability is the same for each s_i except for the two outside levels, where an error can occur in one direction only. Assuming all amplitude levels are equally likely, the average symbol error probability is

$$P_s = \frac{M-1}{M} \Pr \left(|r - s_i| > \frac{\Delta}{2} \right)$$

where Δ is the distance between adjacent signal levels, and also is the distance between adjacent thresholds. For equal amplitude spacing, the amplitudes may be ex-

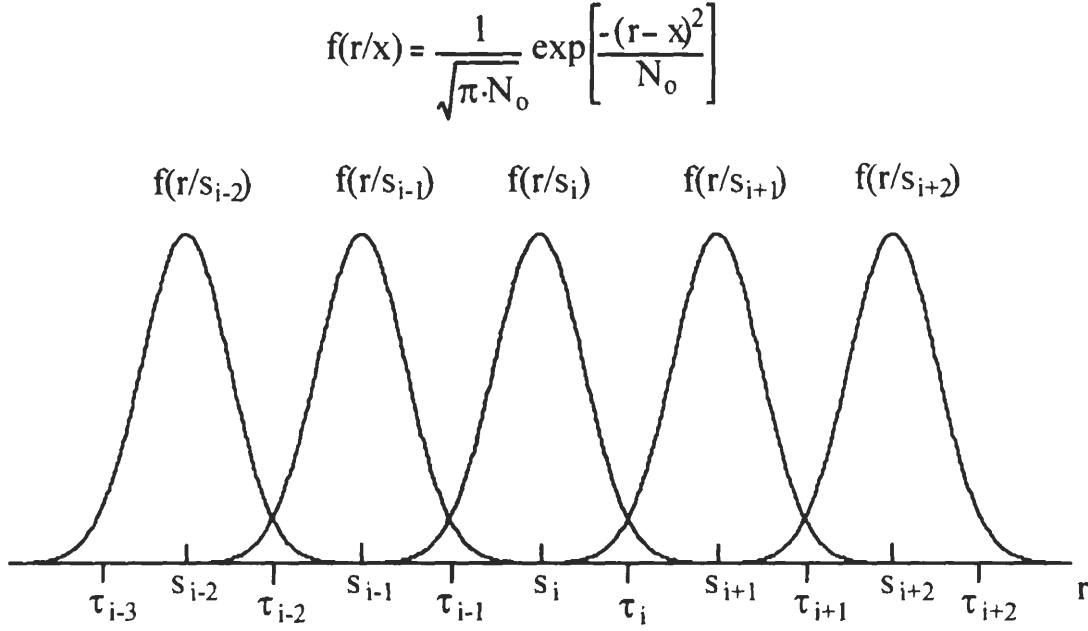


Figure 8.1 Thresholds and decision regions for M-ary PAM.

pressed as

$$s_i = (2i - 1 - M)A_o, \quad i = 1, 2, \dots, M \quad (8.10)$$

where A_o is the smallest amplitude. Then

$$\Delta = |s_i - s_{i-1}| = 2A_o$$

Thus

$$\begin{aligned} P_s &= \frac{M-1}{M} \Pr(|r - s_i| > A_o) = \frac{M-1}{M} \frac{2}{\sqrt{\pi N_o}} \int_{A_o}^{\infty} e^{-x^2/N_o} dx \\ &= \frac{M-1}{M} \frac{2}{\sqrt{2\pi}} \int_{A_o \sqrt{2/N_o}}^{\infty} e^{-x^2/2} dx = \frac{2(M-1)}{M} Q \left(\sqrt{\frac{2A_o^2}{N_o}} \right) \quad (8.11) \end{aligned}$$

The symbol error probability can be expressed in terms of the average energy or power of the signals. The average energy of the signals is

$$E_{avg} = \frac{1}{M} \sum_{i=1}^M E_i = \frac{1}{M} \sum_{i=1}^M s_i^2$$

$$\begin{aligned}
&= \frac{1}{M} \sum_{i=1}^M (2i - 1 - M)^2 A_o^2 = \frac{1}{M} \frac{M(M^2 - 1)A_o^2}{3} \\
&= \frac{1}{3}(M^2 - 1)A_o^2
\end{aligned} \tag{8.12}$$

As a result, (8.11) becomes

$$P_s = \frac{2(M - 1)}{M} Q \left(\sqrt{\frac{6E_{avg}}{(M^2 - 1)N_o}} \right) \tag{8.13}$$

or

$$P_s = \frac{2(M - 1)}{M} Q \left(\sqrt{\frac{6P_{avg}T}{(M^2 - 1)N_o}} \right) \tag{8.14}$$

where $P_{avg} = E_{avg}/T$ is the average power. Since the average energy per bit is $E_{bavg} = E_{avg}/\log_2 M$, (8.13) can be written as

$$P_s = \frac{2(M - 1)}{M} Q \left(\sqrt{\frac{6(\log_2 M)E_{bavg}}{(M^2 - 1)N_o}} \right) \tag{8.15}$$

Figure 8.2 shows the curves of the symbol error probability of the M-ary PAM versus E_{bavg}/N_o . We emphasize that (8.13) to (8.15) depend on only average energy or power of the signal symbols, not the function form of $\phi(t)$. This makes (8.13) to (8.15) applicable to both baseband and bandpass MAM signals. The $M = 2$ case corresponds to NRZ-L (baseband), or BPSK (bandpass), in this case (8.15) reduces to (2.48) or (4.6).

In comparison to MPSK, starting from $M = 4$, the error probability of M-ary PAM is inferior to that of the MPSK. In fact we can easily compare them. From the P_s of the MPSK (4.24) and (8.13), the ratio (MAM over MPSK) of the arguments inside the square root sign of the Q-function is

$$R_M = \frac{3}{(M^2 - 1) \sin^2 \frac{\pi}{M}} \tag{8.16}$$

and is tabulated in Table 8.1. If $M \geq 32$ the degradation is constantly 5.17 dB. This is easily seen from (8.16). For large M , $\sin \pi/M \cong \pi/M$, and $R_M \cong 3/\pi^2 \cong 0.304$. Another fact is that in both MAM and MPSK, for the same error probability, the power increase is 6 dB for doubling M for large M . This can be seen from (4.24) and (8.13). This means in terms of increasing bandwidth efficiency with increased M , both schemes pay the same penalty in BER performance. However, MAM still has a fixed 5.17 dB disadvantage against MPSK. This shows MAM is inferior to MPSK in

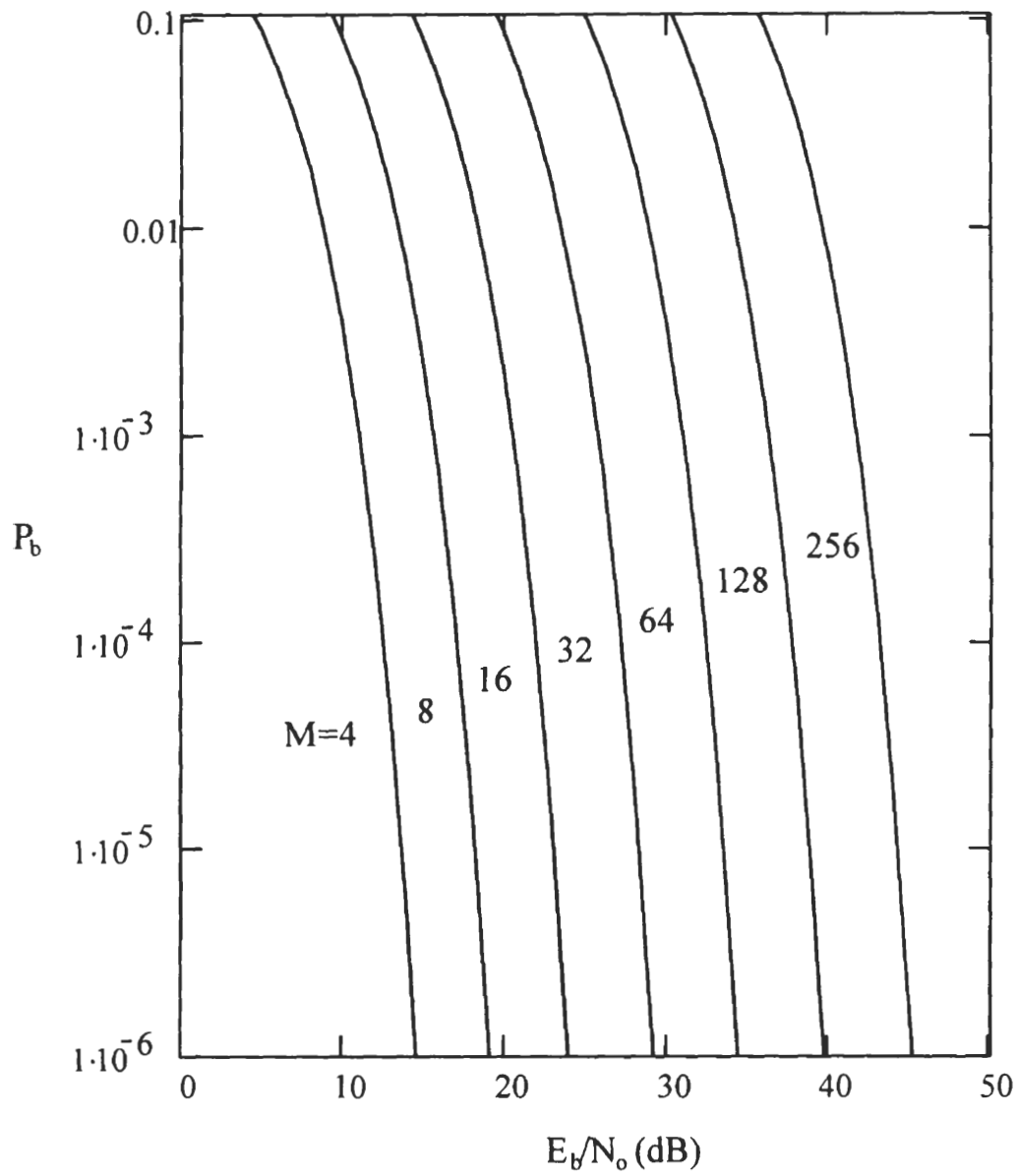


Figure 8.2 Symbol error probability of MAM.

M	R_M	$10 \log R_M$ (dB)
2	1	0
4	0.4	-3.98
8	0.325	-4.88
16	0.309	-5.10
32	0.305	-5.16
64	0.304	-5.17
128	0.304	-5.17
256	0.304	-5.17

Table 8.1 Power penalty of MAM over MPSK.

terms of error probability. However, we will see that going to two-dimensional AM, namely, QAM, improves error rate performance significantly. As a result, QAM is superior to MPSK when $M > 4$.

8.1.3 Modulator and Demodulator for Bandpass MAM

Now we focus on bandpass MAM. The bandpass MAM signal set is

$$s_i(t) = A_i p(t) \cos 2\pi f_c t, \quad 0 \leq t \leq T \quad (8.17)$$

for $i = 1, 2, \dots, M$. It can be easily written in the form of (8.1) by defining $\phi(t) = \sqrt{2/E_p} p(t) \cos 2\pi f_c t$ where E_p is the energy of pulse-shaping signal $p(t)$ in $[0, T]$. Note that $\int_0^T \phi^2(t) dt \cong 1$ for $f_c \gg 1/T$. Thus for most practical cases, $s_i(t) = A_i \sqrt{E_p/2} \phi(t)$ and $s_i = A_i \sqrt{E_p/2}$.

The modulator is shown in Figure 8.3. Figure 8.3(a) is a direct implementation of (8.17). The level generator takes $n = \log_2 M$ bits from the binary data stream and maps them into an amplitude level $A_k \in \{A_i\}$, where the subscript k indicates the k th symbol interval. The mapping is preferably Gray coding so that the n -tuples representing the adjacent amplitudes differ only by one bit. The functions of the rest blocks are self-explanatory. The equivalent implementation is shown in Figure 8.3(b). It is more practical for hardware implementation. The $p(t)$ multiplier is replaced with a filter with an impulse response $p(t)$. In order to generate a pulse $A_i p(t)$, the input to the filter must be an impulse $A_i \delta(t)$. In practice, this can be realized by a very narrow pulse with amplitude A_i .

The optimum receiver implementing the minimum distance rule is shown in Figure 8.4 where the last block is a threshold detector. Figure 8.4(a) is a direct implementation of (8.9) and the minimum distance rule. In Figure 8.4(b), the correlation with $p(t)$ is replaced by a matched filter and a sampler. Figure 8.4(b) is more practical for

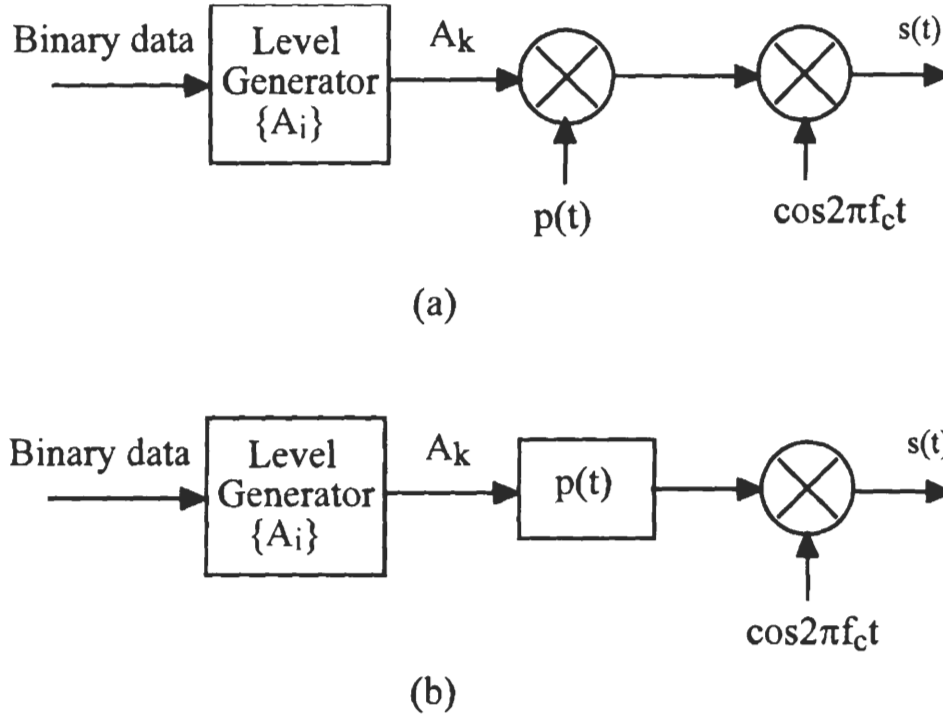


Figure 8.3 Bandpass MAM modulator.

hardware implementation. The equivalence between parts (a) and (b) in Figure 8.4 can be shown as follows. For the signal part in the noise-corrupted received signal, the output of the down-converter in both cases is

$$\begin{aligned}
 & A_i p(t) \cos 2\pi f_c t \sqrt{\frac{2}{E_p}} \cos 2\pi f_c t \\
 &= \frac{1}{2} A_i \sqrt{\frac{2}{E_p}} p(t) [1 + \cos 4\pi f_c t]
 \end{aligned}$$

In Figure 8.4(a), the integrator output is

$$\frac{1}{2} A_i \sqrt{\frac{2}{E_p}} \int_0^T p^2(t) [1 + \cos 4\pi f_c t] dt = A_i \sqrt{\frac{E_p}{2}} = s_i$$

for $f_c \gg 1/T$. In Figure 8.4(b), the high-frequency term $\cos 4\pi f_c t$ is blocked by the low-pass filter $p(T - t)$. The output of the filter is

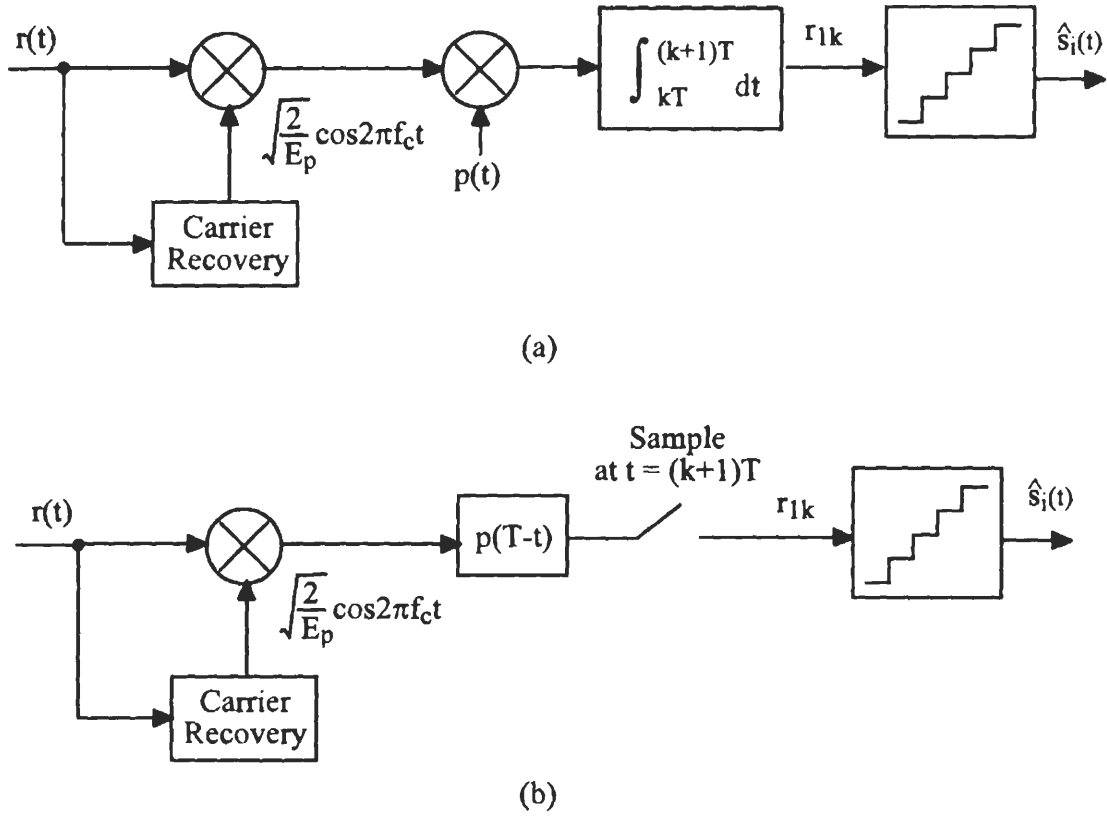


Figure 8.4 Optimum coherent demodulators for bandpass MAM.

$$\begin{aligned}
 & \frac{1}{2} A_i \sqrt{\frac{2}{E_p}} \int_0^t p(\tau) p(T - t + \tau) d\tau \Big|_{t=T} \\
 &= \frac{1}{2} A_i \sqrt{\frac{2}{E_p}} \int_0^T p(\tau) p(\tau) d\tau = A_i \sqrt{\frac{E_p}{2}} = s_i
 \end{aligned}$$

which is exactly the same as the output of the integrator in Figure 8.4(a). As to the noise in the received signal, parts (a) and (b) in Figure 8.4 are also equivalent. Assuming the noise in the received signal is $n(t)$. The output noise of the integrator in Figure 8.4(a) is $\int_0^T n(t) \phi(t) dt$. The output noise of the down-converter is

$$n_1(t) = n(t) \sqrt{\frac{2}{E_p}} \cos 2\pi f_c t = \frac{n(t) \phi(t)}{p(t)}$$

Then the output of the matched filter is

$$\begin{aligned}
 \int_0^t n_1(\tau) p(T - t + \tau) d\tau \Big|_{t=T} &= \int_0^T \frac{n(\tau) \phi(\tau)}{p(\tau)} p(T - t + \tau) d\tau \Big|_{t=T} \\
 &= \int_0^T \frac{n(\tau) \phi(\tau)}{p(\tau)} p(\tau) d\tau \Big|_{t=T} \\
 &= \int_0^T n(\tau) \phi(\tau) d\tau
 \end{aligned}$$

which is exactly the same as that of the correlator. Thus the equivalence between parts (a) and (b) in Figure 8.4 is established.

The threshold detector has $M - 1$ thresholds, each placed in the middle of two signal points (refer to Figure 8.1). The correlator computes the r and the threshold detector is actually computing the distance of r to the amplitudes s_i , $i = 1, 2, \dots, M$ and chooses the smallest. Note that the reference signal can be any scaled version of $\phi(t)$ as long as the thresholds are scaled accordingly.

The coherent carrier in the coherent demodulator can be generated using the synchronization methods described in Chapter 4, such as a squaring loop (Figure 4.35 with $M = 2$). The π phase ambiguity associated with the squaring loop can be solved by differential coding. The Costas loop (Figure 4.36) can also be used to demodulate MAM. The symbol timing clock can be recovered by using the methods described in Chapter 4.

8.1.4 On-Off Keying

The simplest MAM, is the OOK, which we mentioned in Chapter 1. The OOK signal set is

$$\begin{aligned}
 s_1(t) &= A \cos 2\pi f_c t, & \text{for } a = 1, & \quad 0 \leq t \leq T \\
 s_2(t) &= 0, & \text{for } a = 0, & \quad 0 \leq t \leq T
 \end{aligned} \tag{8.18}$$

where a is the binary data which are assumed uncorrelated and equally likely. The complex envelope of the OOK signal $s(t)$ on the entire time axis is

$$\tilde{s}(t) = \sum_{k=-\infty}^{\infty} A_k p(t - kT), \quad -\infty < t < \infty$$

where $A_k \in \{0, A\}$, $p(t)$ is a rectangular pulse with unit amplitude whose $P(f)$ is given by (8.7). Since $m_A = A/2$ and $\sigma_A^2 = A^2/4$, $P(\frac{k}{T}) = 0$ except for $k = 0$,

from (8.5) we have the PSD for OOK as

$$\Psi_{\tilde{s}}(f) = \frac{A^2 T}{4} \left(\frac{\sin \pi f T}{\pi f T} \right)^2 + \frac{A^2}{4} \delta(f) \quad (8.19)$$

Note this is exactly the same as the PSD of the unipolar NRZ line codes (see (2.25) and Figure 2.3(b)).

The symbol error probability for coherent demodulation of OOK can be obtained from (8.13) with $E_{avg} = E_b$, or directly from (B.32), the BER expression for binary signaling. That is

$$P_b = Q \left(\sqrt{\frac{E_b}{2N_o}} \right) \quad (8.20)$$

where E_b is the average bit energy. When compared with BPSK, the PSD is the same except that it has a spectral line at f_c , which can be locked on by a phase lock loop to recover the carrier, but the BER performance of OOK is 3 dB inferior to that of BPSK. OOK is not usually preferred against BPSK.

8.2 QAM SIGNAL DESCRIPTION

Having studied MAM, we are ready to discuss QAM. In MAM schemes, signals have the same phase but different amplitudes. In MPSK schemes, signals have the same amplitude but different phases. Naturally, the next step of development is to consider using both amplitude and phase modulations in a scheme (QAM). That is

$$s_i(t) = A_i \cos(2\pi f_c t + \theta_i), \quad i = 1, 2, \dots, M \quad (8.21)$$

where A_i is the amplitude and θ_i is the phase of the i th signal in the M-ary signal set. Pulse shaping is usually used to improve the spectrum and for ISI control purpose in QAM. With pulse shaping, the QAM signal is

$$s_i(t) = A_i p(t) \cos(2\pi f_c t + \theta_i), \quad i = 1, 2, \dots, M \quad (8.22)$$

where $p(t)$ is a smooth pulse defined on $[0, T]$.¹ Expression (8.22) can be written as

$$s_i(t) = A_{i1}p(t) \cos 2\pi f_c t - A_{i2}p(t) \sin 2\pi f_c t \quad (8.23)$$

where

$$A_{i1} = A_i \cos \theta_i \quad (8.24)$$

$$A_{i2} = A_i \sin \theta_i \quad (8.25)$$

and

$$A_i = \sqrt{A_{i1}^2 + A_{i2}^2} \quad (8.26)$$

Similar to MPSK, QAM signal can be expressed as a linear combination of two orthonormal functions. Expression (8.23) can be written as

$$s_i(t) = s_{i1}\phi_1(t) + s_{i2}\phi_2(t) \quad (8.27)$$

where

$$\phi_1(t) = \sqrt{\frac{2}{E_p}} p(t) \cos 2\pi f_c t, \quad 0 \leq t \leq T \quad (8.28)$$

$$\phi_2(t) = -\sqrt{\frac{2}{E_p}} p(t) \sin 2\pi f_c t, \quad 0 \leq t \leq T \quad (8.29)$$

and

$$s_{i1} = \sqrt{\frac{E_p}{2}} A_{i1} = \sqrt{\frac{E_p}{2}} A_i \cos \theta_i \quad (8.30)$$

$$s_{i2} = \sqrt{\frac{E_p}{2}} A_{i2} = \sqrt{\frac{E_p}{2}} A_i \sin \theta_i \quad (8.31)$$

where E_p is the energy of $p(t)$ in $[0, T]$. That is $E_p = \int_0^T p^2(t) dt$. The factor $\sqrt{2/E_p}$ is to normalize the basis functions $\phi_1(t)$ and $\phi_2(t)$.

¹ Even if pulse shaping is not desired, there is still inevitably pulse shaping due to the limited bandwidth of the system. In fact, deliberate pulse shaping is usually achieved through filtering. That is to make $P(f) = H_T(f)H_C(f)H_R(f)$ or equivalently $p(t) = h_T(t) * h_C(t) * h_R(t)$, where $h_T(t)$, $h_C(t)$ and $h_R(t)$ are the impulse responses of the transmitter filter, channel, and receiver filter, respectively. $H_T(f)$, $H_C(f)$, and $H_R(f)$ are their transfer functions. A common choice of $P(f)$ is the raised-cosine, whose time domain function $p(t)$ has zero values at sampling instants except for at $t = 0$. Thus $p(t)$ incurs no ISI. However, the raised-cosine response is noncausal, only an approximate delayed version is realizable. See [1, pp. 100-102].

It can be easily verified that the basis functions $\phi_1(t)$ and $\phi_2(t)$ are virtually orthonormal for $f_c \gg 1/T$. When $f_c \gg 1/T$, $p(t)$ is a slow-varying envelope. First they are virtually normalized since

$$\begin{aligned} \int_0^T \phi_1^2(t) dt &= \frac{2}{E_p} \int_0^T p^2(t) \cos^2 2\pi f_c t dt \\ &= \frac{1}{E_p} \int_0^T p^2(t) [1 + \cos 4\pi f_c t] dt \\ &\cong 1, \quad \text{for } f_c \gg 1/T \end{aligned}$$

The same is true for $\phi_2(t)$. Second, they are virtually orthogonal since

$$\begin{aligned} \int_0^T \phi_1(t) \phi_2(t) dt &= -\frac{2}{E_p} \int_0^T p^2(t) \cos 2\pi f_c t \sin 2\pi f_c t dt \\ &= -\frac{2}{E_p} \int_0^T p^2(t) \sin 4\pi f_c t dt \\ &\cong 0, \quad \text{for } f_c \gg 1/T \end{aligned}$$

Thus for most practical cases, $\phi_1(t)$ and $\phi_2(t)$ are orthonormal. When there is no pulse shaping, that is, $p(t) = 1$ in $[0, T]$, $E_p = T$. Then (8.28) and (8.29) have the same forms of (4.2) and (4.3). They are precisely orthonormal.

The energy of the i th signal is

$$E_i = \int_0^T s_i^2(t) dt \cong \frac{1}{2} A_i^2 E_p \quad (8.32)$$

and the average signal energy is

$$E_{avg} = \frac{1}{2} E_p \cdot E\{A_i^2\} \quad (8.33)$$

The average power is

$$P_{avg} = \frac{E_{avg}}{T} \quad (8.34)$$

The average amplitude is

$$A_{avg} = \sqrt{2P_{avg}} \quad (8.35)$$

Similar to MPSK, a geometric representation called constellation is a very clear way of describing a QAM signal set. The horizontal axis of the constellation plane is $\phi_1(t)$ and the vertical axis is $\phi_2(t)$. A QAM signal is represented by a point (or

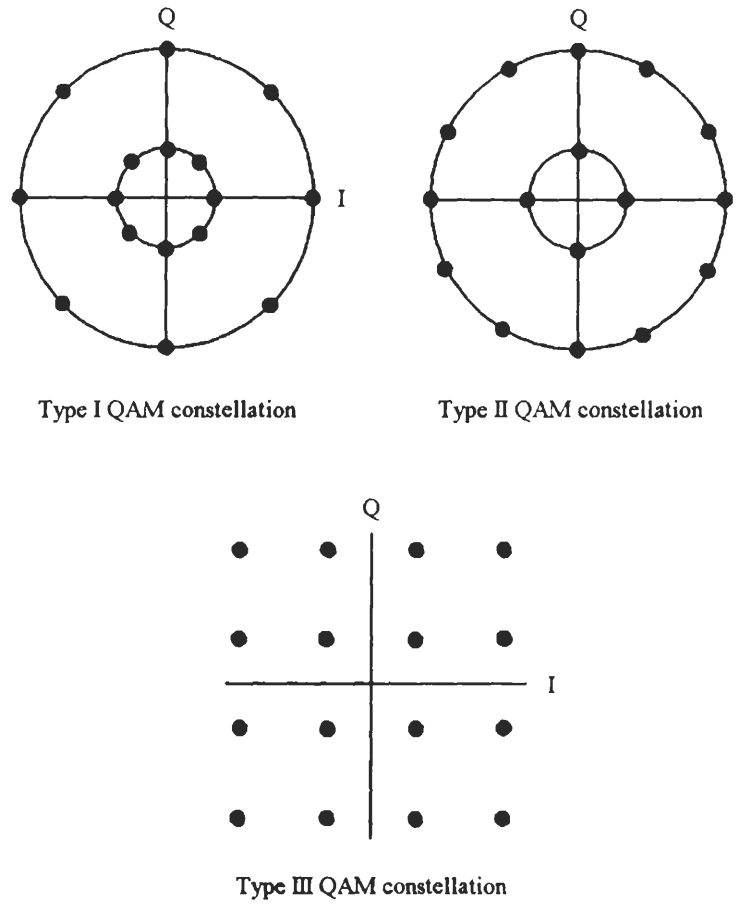


Figure 8.5 Examples of type I, II and III QAM constellations. From [8]. Copyright © 1994 IEEE.

vector, or phasor) with coordinates (s_{i1}, s_{i2}) . Alternatively, the two axes can be simply chosen as $p(t) \cos 2\pi f_c t$ and $-p(t) \sin 2\pi f_c t$. Then the signal coordinates are (A_{i1}, A_{i2}) . The two axes sometimes are simply labeled as I-axis and Q-axis, and sometimes are even left unlabeled. Figure 8.5 shows examples of three types of QAM constellations.

Now let us examine the properties of the QAM constellation. Assuming the axes are $\phi_1(t)$ and $\phi_2(t)$, then each signal is represented by the phasor

$$\mathbf{s}_i = (s_{i1}, s_{i2})$$

The magnitude of the phasor is

$$||\mathbf{s}_i|| = \sqrt{s_{i1}^2 + s_{i2}^2} = \sqrt{E_i} \quad (8.36)$$

which is related to the signal amplitude by (from (8.32) and (8.36))

$$A_i = \sqrt{\frac{2}{E_p}} ||\mathbf{s}_i|| \quad (8.37)$$

The average energy is

$$E_{avg} = E\{E_i\} = E\{||\mathbf{s}_i||^2\} \quad (8.38)$$

The phase θ_i is the angle of the corresponding phasor

$$\theta_i = \tan^{-1} \frac{s_{i2}}{s_{i1}} \quad (8.39)$$

The distance between any pair of phasors is

$$\begin{aligned} d_{ij} &= \sqrt{|\mathbf{s}_i - \mathbf{s}_j|^2} \\ &= \sqrt{(s_{i1} - s_{j1})^2 + (s_{i2} - s_{j2})^2}, \quad i, j = 1, 2, \dots, M \end{aligned} \quad (8.40)$$

Depending what values (s_{i1}, s_{i2}) or (A_i, θ_i) are assigned with, a variety of QAM constellations can be realized.

8.3 QAM CONSTELLATIONS

The first QAM scheme was proposed by C. R. Cahn in 1960 [2]. He simply extended phase modulation to a multi-amplitude phase modulation. That is, there is more than one amplitude associated with an allowed phase. In the constellation, a fixed number of signal points (or phasors) are equally spaced on each of the N circles, where N is the number of amplitude levels (Figure 8.5(a)). This is called type I constellation in the literature. In a type I constellation, the points on the inner ring are closest together in distance and are most vulnerable to errors. To overcome this problem, type II constellation was proposed by Hancock and Lucky a few months later [3](Figure 8.5(b)). In a type II constellation, signal points are still on circles, but the number of points on the inner circle is less than the number of points on the outer circle, making the distance between two adjacent points on the inner circle approximately equal to that on the outer circle. Type III constellation is the square

QAM constellation shown in Figure 8.5(c), which was proposed by Campopiano and Glazer in 1962 [4]. Their analysis showed that the type III system offered a very small improvement in performance over the type II system, but its implementation would be considerably simpler than that of type I and II. Due to this, the type III constellation has been the most widely used system. Some other two dimensional constellations considered in the literature are given in Figure 8.6. The circular constellations are denoted by the notation $(n_1, n_2, \dots)$ where n_1 is the number of signal points on the inner circle, n_2 is the number of signal points on the next circle, and so on. Figure 8.6 contains the type II and type III constellations.

When designing a constellation, consideration must be given to:

1. The minimum Euclidean distance $d_{\min}$ among the phasors (signal points). It should be as large as possible under other constraints, since it determines the symbol error probability of the modulation scheme.
2. The phase differences among the phasors. It should be as large as possible under other constraints, since it determines the phase jitter immunity and hence the scheme's resilience against the carrier- and clock-recovery imperfections and channel phase rotations.
3. The average power of the phasors. It should be as small as possible under other constraints.
4. The ratio of the peak-to-average phasor power, which is a measure of robustness against nonlinear distortion caused by the power amplifier. It should be as close to unity as possible under other constraints.
5. The implementation complexity.
6. Other properties, such as resilience against fading.

Research results have shown that the square constellation (type III) is the most appropriate choice in AWGN channels. It can be easily generated as two MAM signals impressed on two phase-quadrature carriers. It can be easily demodulated to yield two quadrature components. Each component can be individually detected by comparing it to a set of thresholds. A few of the other constellations offer slightly better error performance, but with a much more complicated system implementation. Therefore we will concentrate on the square constellation in this chapter. The type I constellation (also called star constellation) is not optimum in terms of $d_{\min}$ under the constraint of average phasor power. However, it allows efficient differential encoding and decoding methods to be used. This makes it suitable for fading channels. Its application in fading channels will be covered in Chapter 10.

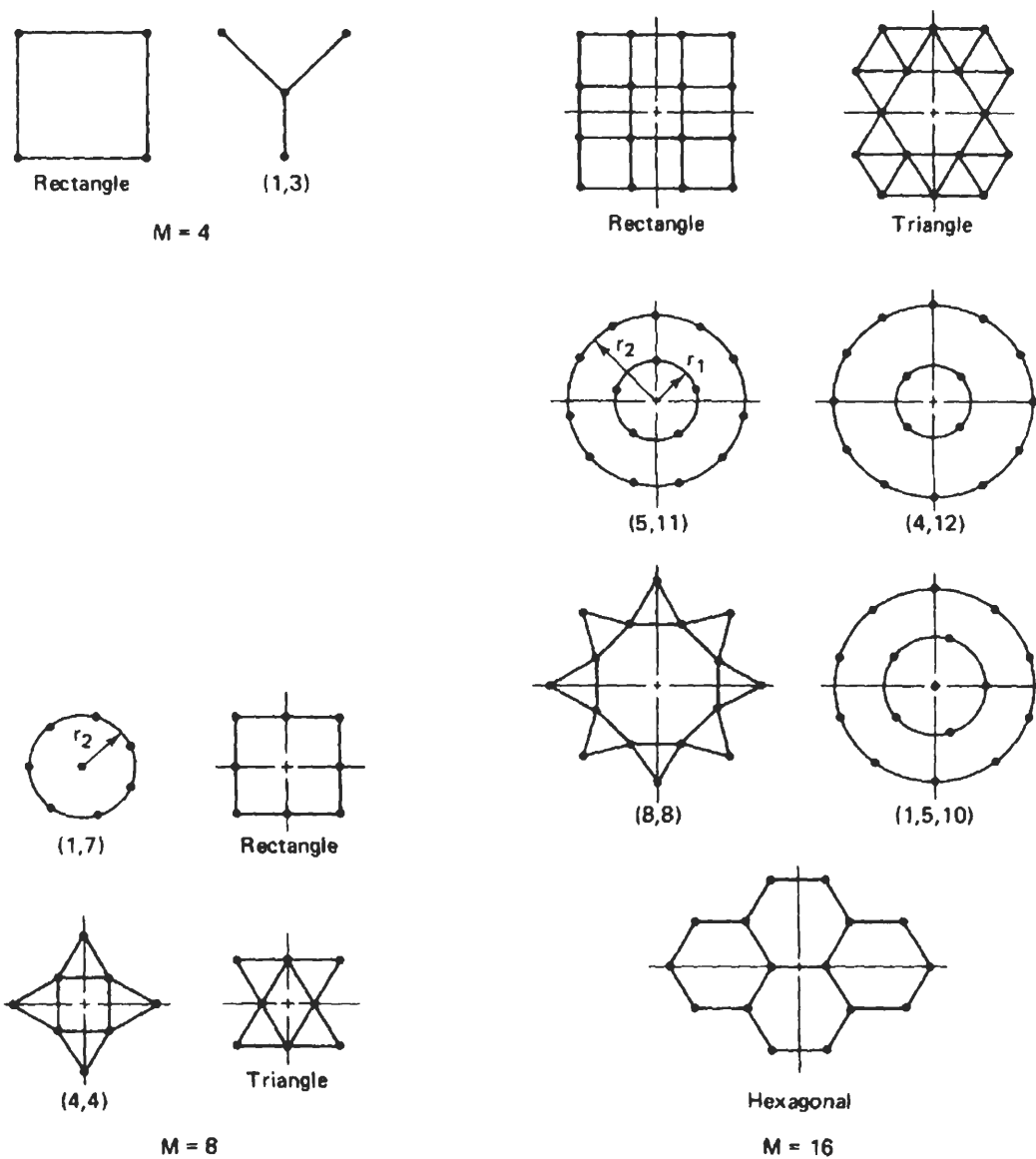


Figure 8.6 Various QAM constellations. From [5]. Copyright © 1974 IEEE.

8.3.1 Square QAM

For M-ary square QAM signals, (8.23) and (8.27) can be written in the following form

$$\begin{aligned} s_i(t) &= I_i \sqrt{\frac{E_0}{E_p}} p(t) \cos 2\pi f_c t - Q_i \sqrt{\frac{E_0}{E_p}} p(t) \sin 2\pi f_c t \\ &= I_i \sqrt{\frac{E_0}{2}} \phi_1(t) + Q_i \sqrt{\frac{E_0}{2}} \phi_2(t) \end{aligned} \quad (8.41)$$

where E_0 is the energy of the signal with the lowest amplitude, and (I_i, Q_i) are a pair of independent integers which determine the location of the signal point in the constellation. The minimum values of (I_i, Q_i) are $(\pm 1, \pm 1)$. The pair (I_i, Q_i) is an element of the $L \times L$ matrix:

$$[I_i, Q_i] = \begin{bmatrix} (-L+1, L-1) & (-L+3, L-1) & \cdots & (L-1, L-1) \\ (-L+1, L-3) & (-L+3, L-3) & \cdots & (L-1, L-3) \\ \vdots & \vdots & & \vdots \\ (-L+1, -L+1) & (-L+3, -L+1) & \cdots & (L-1, -L+1) \end{bmatrix} \quad (8.42)$$

where

$$L = \sqrt{M}, \quad M = 4^n, \quad n = 1, 2, 3, \dots$$

For example, for the 16-QAM in Figure 8.7, where $L = 4$, the matrix is

$$[I_i, Q_i] = \begin{bmatrix} (-3, 3) & (-1, 3) & (1, 3) & (3, 3) \\ (-3, 1) & (-1, 1) & (1, 1) & (3, 1) \\ (-3, -1) & (-1, -1) & (1, -1) & (3, -1) \\ (-3, -3) & (-1, -3) & (1, -3) & (3, -3) \end{bmatrix} \quad (8.43)$$

When $M = 2^n$ but not 4^n , L is not an integer, we cannot use the matrix (8.42) directly to define the QAM. However, we may use a modified matrix to define the QAM. For example, the 32-QAM can be defined by a 6×6 matrix without the four elements on the four corners.

The constellation can be conveniently expressed in terms of (I_i, Q_i) . The phasors for the square QAM are

$$\mathbf{s}_i = \left(I_i \sqrt{\frac{E_0}{2}}, Q_i \sqrt{\frac{E_0}{2}} \right) \quad i = 1, 2, \dots, M$$

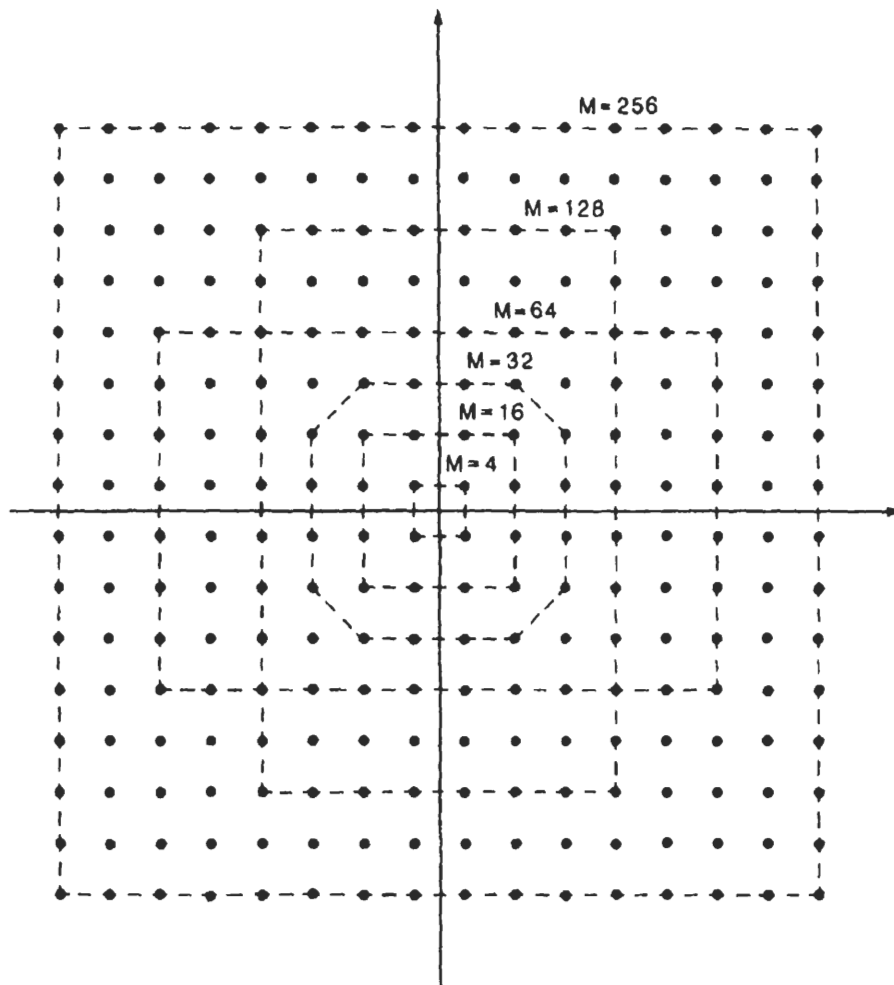


Figure 8.7 Square QAM constellations. From [6, p.224]. Copyright © 1987 Prentice Hall.

The magnitude of a phasor is

$$||\mathbf{s}_i|| = \sqrt{\frac{E_0}{2}(I_i^2 + Q_i^2)}$$

The QAM signal in (8.41) can also be written as

$$s_i(t) = A_i \cos(2\pi f_c t + \theta_i)$$

where the amplitude is

$$A_i = \sqrt{\frac{E_0}{E_p}(I_i^2 + Q_i^2)} = \sqrt{\frac{2}{E_p}} ||\mathbf{s}_i||$$

The phase θ_i is the angle of the corresponding phasor

$$\theta_i = \tan^{-1} \frac{Q_i}{I_i}$$

The distance between any pair of phasors is

$$\begin{aligned} d_{ij} &= \sqrt{|\mathbf{s}_i - \mathbf{s}_j|^2} \\ &= \sqrt{\frac{E_0}{2}[(I_i - I_j)^2 + (Q_i - Q_j)^2]}, \quad i, j = 1, 2, \dots, M \end{aligned}$$

The average energy is

$$E_{avg} = E\left\{\frac{E_0}{2}(I_i^2 + Q_i^2)\right\} = \frac{E_0}{2}[E\{I_i^2\} + E\{Q_i^2\}] = E_0 E\{I_i^2\}$$

and the average power is

$$P_{avg} = \frac{E_0}{T} E\{I_i^2\}$$

where for the strict square ($L \times L$) QAM

$$\begin{aligned} E\{I_i^2\} &= \frac{1}{L}[(-(L-1))^2 + (-(L-3))^2 + \dots + (L-3)^2 + (L-1)^2] \\ &= \frac{2}{L}[1^2 + 3^2 + \dots + (L-1)^2] \\ &= \frac{1}{3}(L^2 - 1) = \frac{1}{3}(M - 1) \end{aligned} \tag{8.44}$$

Thus

$$P_{avg} = \frac{E_0}{3T}(M-1) = P_0 \frac{1}{3}(M-1) \quad (8.45)$$

where P_0 is the power of the smallest signal.

8.4 POWER SPECTRAL DENSITY

Using the PSD formula for quadrature modulation (see (A.21) in Appendix A), the PSD of the square QAM can be computed as follows.

In order to include the most general case, we consider QAM with pulse shaping.

$$s_i(t) = A_i p(t) \cos(2\pi f_c t + \theta_i), \quad i = 1, 2, \dots, M$$

On the entire time axis, the QAM signal can be written as

$$s(t) = \text{Re} \left\{ \left[\sum_{k=-\infty}^{\infty} A_k \exp(j\theta_k) p(t - kT) \right] \exp(j2\pi f_c t) \right\}, \quad -\infty < t < \infty \quad (8.46)$$

The complex envelope of the QAM signal is

$$\begin{aligned} \tilde{s}(t) &= \sum_{k=-\infty}^{\infty} A_k \exp(j\theta_k) p(t - kT) \\ &= \sum_{k=-\infty}^{\infty} A_k \cos \theta_k p(t - kT) + j \sum_{k=-\infty}^{\infty} A_k \sin \theta_k p(t - kT) \quad (8.47) \\ &= \sum_{k=-\infty}^{\infty} A_{k1} p(t - kT) + j \sum_{k=-\infty}^{\infty} A_{k2} p(t - kT) \end{aligned}$$

where

$$A_{k1} = A_k \cos \theta_k$$

$$A_{k2} = A_k \sin \theta_k$$

These are random variables with equal probability for each value. They have zero means for symmetrical constellations (Figures 8.5 to 8.7). The variances of them depend on the constellation shape.

$$\sigma_1^2 = E\{A_{k1}^2\}$$

$$\sigma_2^2 = E\{A_{k2}^2\}$$

From (8.38), the average power of the signal is

$$P_{avg} = \frac{1}{2T} E_p \cdot E\{A_i^2\} = \frac{1}{2T} E_p (\sigma_1^2 + \sigma_2^2)$$

Now we use the PSD formula (A.21) in Appendix A to compute the PSD of QAM. We rewrite it in the following

$$\Psi_{\tilde{s}}(f) = \frac{\sigma_x^2 |P(f)|^2}{T} + \frac{\sigma_y^2 |Q(f)|^2}{T}$$

where $P(f)$ and $Q(f)$ are the spectra of the I-channel signal and Q-channel signal pulse shape, respectively. This is a very general formula, applicable to any quadrature modulated signal. For QAM schemes $P(f) = Q(f)$, $\sigma_x^2 = \sigma_1^2$ and $\sigma_y^2 = \sigma_2^2$, we have

$$\Psi_{\tilde{s}}(f) = \frac{|P(f)|^2}{T} (\sigma_1^2 + \sigma_2^2) = \frac{2P_{avg}}{E_p} |P(f)|^2 \quad (8.48)$$

This equation tells us that the shape of the PSD of a QAM scheme is determined by the baseband pulse shape, and the magnitude of the PSD is determined by the average power (or average amplitude) of the QAM signal set. It is also worthwhile to point out that the shape of the PSD of a QAM scheme is independent of the constellation. In other words, no matter what the constellation is, be it square, circular or others, the PSD shape is the same as long as the $p(t)$ is the same, the PSD magnitude is also the same as long as the average signal power is also the same.

Without particular pulse shaping, $p(t)$ is just a rectangular pulse with unit amplitude. Then $E_p = T$ and

$$|P(f)| = \left| T \frac{\sin \pi f T}{\pi f T} \right|$$

Therefore

$$\begin{aligned} \Psi_{\tilde{s}}(f) &= 2P_{avg} T \left(\frac{\sin \pi f T}{\pi f T} \right)^2 \\ &= A_{avg}^2 T \left(\frac{\sin \pi f T}{\pi f T} \right)^2 \\ &= A_{avg}^2 n T_b \left(\frac{\sin \pi f n T_b}{\pi f n T_b} \right)^2 \end{aligned} \quad (8.49)$$

where $n = \log_2 M$ and $T_b = T/n$ is the bit period. This PSD has the same shape of the PSD of MPSK (see (4.26)). The only difference lies in the magnitude. In the MPSK case, the PSD magnitude depends on the signal amplitude because there is only one signal amplitude. In the QAM case, the PSD magnitude depends on the average signal amplitude. Thus, the PSD curves for MPSK in Figure 4.15 are also applicable to QAM schemes as long as the average amplitude of the QAM is used. For example, for M-ary square QAM, from (8.49) and (8.45) we have

$$\begin{aligned}\Psi_{\bar{s}}(f) &= \frac{2E_0}{3}(M-1) \left(\frac{\sin \pi f T}{\pi f T} \right)^2 \\ &= \frac{2E_0}{3}(M-1) \left(\frac{\sin \pi f n T_b}{\pi f n T_b} \right)^2\end{aligned}\quad (8.50)$$

8.5 MODULATOR

The QAM modulator is almost identical to that of MPSK since both of them are quadrature schemes. We can write the QAM signal as

$$s(t) = s_1(t) \cos 2\pi f_c t - s_2(t) \sin 2\pi f_c t, \quad -\infty < t < \infty \quad (8.51)$$

where

$$\begin{aligned}s_1(t) &= \sum_{k=-\infty}^{\infty} A_{k1} p(t - kT) \\ s_2(t) &= \sum_{k=-\infty}^{\infty} A_{k2} p(t - kT)\end{aligned}$$

The modulator derived directly from (8.51) is shown in Figure 8.8. If pulse shaping is not desired, the $p(t)$ block will be absent. The data bit sequence is divided into n -tuples of n bits. There are $M = 2^n$ distinct n -tuples. Each n -tuple of the input bits is used to control the level generator. The level generator provides the I- and Q-channel the particular sign and level for a signal's horizontal and vertical coordinates (A_{k1}, A_{k2}) , respectively. The mapping from n -tuples to QAM points are usually Gray coded for minimizing bit errors. For square QAM, perfect Gray coding is possible. Figure 8.9 is a Gray coded square 16-QAM constellation. For some constellations, such as a circular QAM with four points on the inner ring and eight on the outer, it is not possible to have perfect Gray coding.

Digital synthesis techniques can be used to generate QAM signals. Each signal

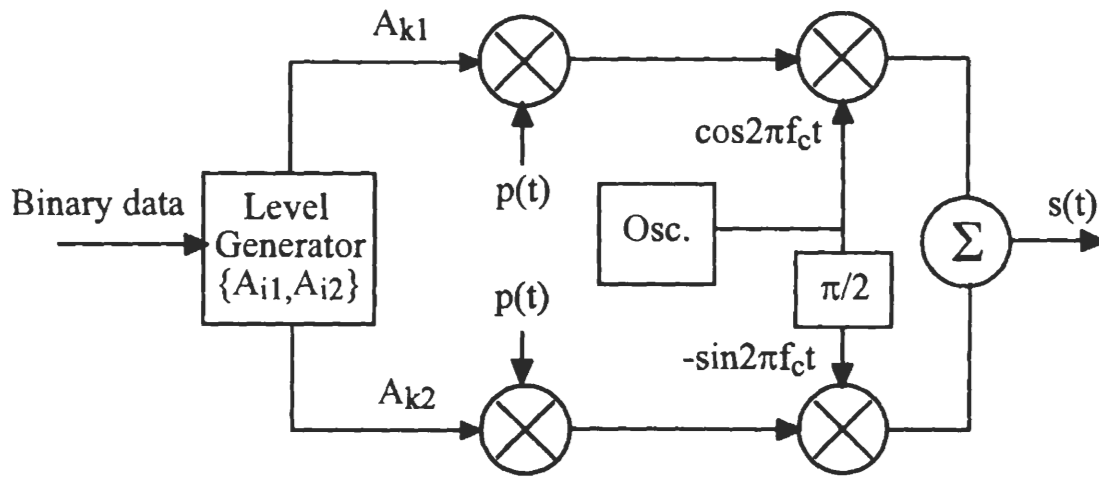


Figure 8.8 QAM modulator.

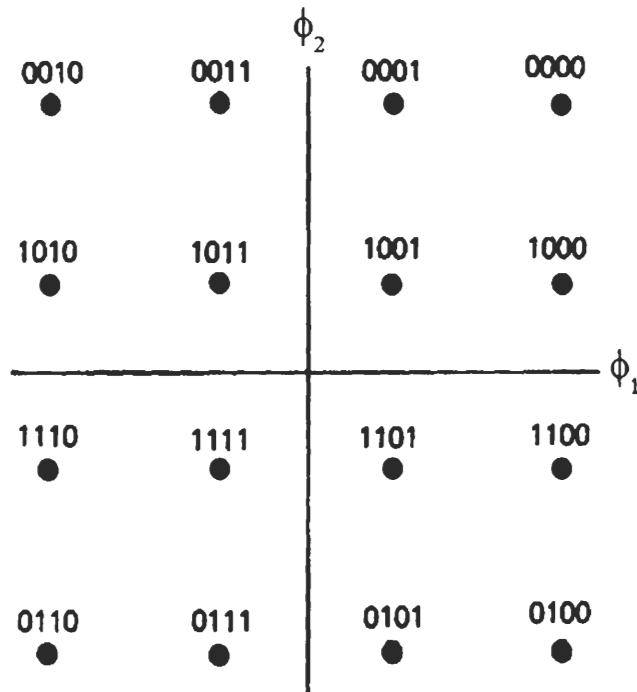


Figure 8.9 Gray coded square 16-QAM constellation. The signal points are labeled with 4-bit Gray codes.

in the constellation can be stored as a set of samples and the data n -tuple is used as the address to obtain the samples. The samples are fed to a D/A converter whose output is the desired QAM signal.

8.6 DEMODULATOR

Similar to MPSK, the coherent demodulation of QAM could be implemented by one of the coherent detectors for M -ary signals as described in Appendix B. Since the QAM signal set has only two basis functions, the simplest receiver is the one that uses two correlators (Figure B.8 with $N = 2$). Due to the special characteristic of the QAM signal, the general demodulator of Figure B.8 can be simplified.

The received signal is

$$r(t) = s_i(t) + n(t)$$

According to (B.37) in Appendix B, for QAM signal detection the sufficient statistic is the (squared) distance²

$$l_i = (r_1 - s_{i1})^2 + (r_2 - s_{i2})^2 \quad (8.52)$$

where

$$r_1 \triangleq \int_0^T r(t)\phi_1(t)dt = s_{i1} + n_1$$

$$r_2 \triangleq \int_0^T r(t)\phi_2(t)dt = s_{i2} + n_2$$

are independent Gaussian random variables with mean values s_{i1} and s_{i2} , respectively. Their variance is $N_o/2$. The pair (r_1, r_2) determines a point in the QAM constellation plane, representing the received noisy signal. The detector compares the distances from (r_1, r_2) to all pairs of (s_{i1}, s_{i2}) and chooses the closest one.

Figure 8.10 is the demodulator based on the above decision rule where subscript k indicates the k th symbol period. Note that the amplitude of the reference signals can be any value, which is $\sqrt{2/E_p}$ in the figure, as long as (s_{i1}, s_{i2}) are also scaled accordingly. As we have shown in Section 8.1.3, the integrators may be replaced by matched filters whose outputs are sampled at $t = (k+1)T$ (Figure 8.11). The filter impulse responses match to the shaping pulse $p(t)$. For the square QAM, the r_{1k} and

² Note that this is different from the sufficient statistic of MPSK. In the case of MPSK, each signal has the same energy, thus (B.38) with B_j dropped instead of (B.37) is used, which leads to a decision rule that compares only angles not distances.

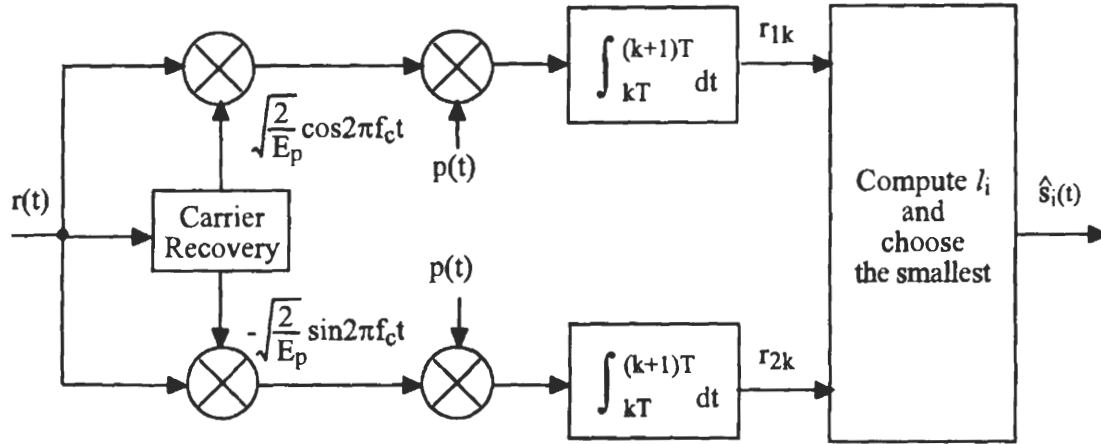


Figure 8.10 Coherent demodulator for QAM.

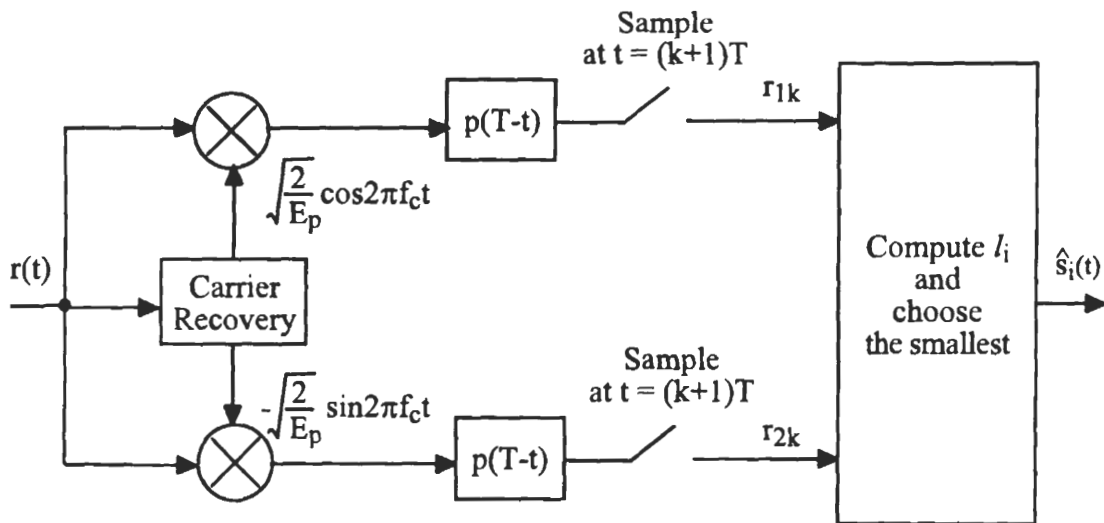


Figure 8.11 Coherent QAM demodulator using matched filters.

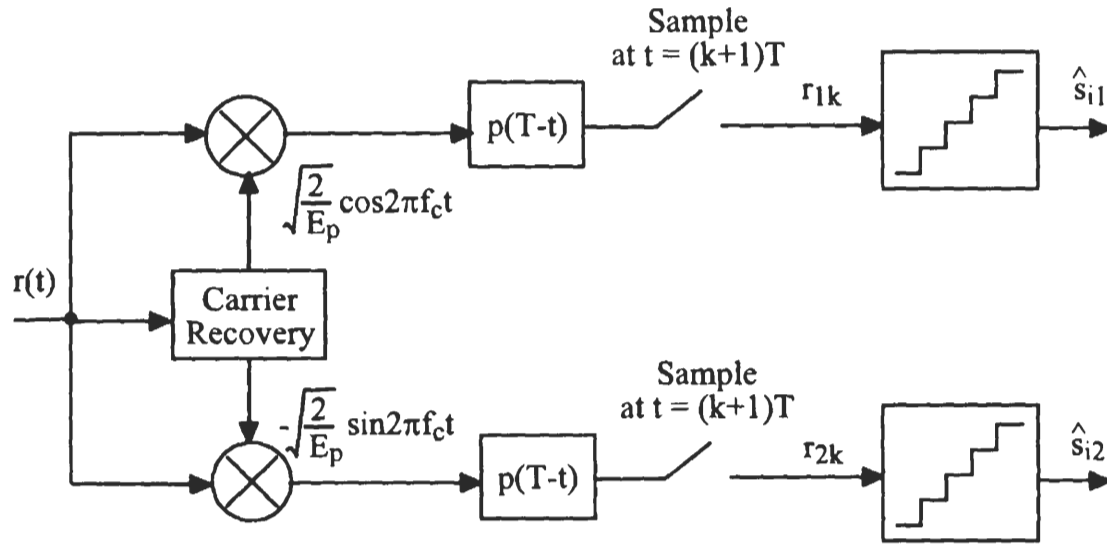


Figure 8.12 Coherent demodulator for square QAM using threshold detectors. I-channel and Q-channel are demodulated separately.

r_{2k} can be detected separately by two multi-threshold detectors to yield s_{i1} and s_{i2} , and then signal $s_i(t)$ can be determined (Figure 8.12).

8.7 ERROR PROBABILITY

For square QAM constellations with $M = 2^k$ where k is even, the QAM constellation is equivalent to two MAM signals on quadrature carriers, each having $L = \sqrt{M}$ signal points. As we have seen from the last section, each MAM signal can be demodulated separately. A QAM symbol is detected correctly only when two MAM symbols are detected correctly. Thus the probability of correct detection of a QAM symbol is

$$P_c = (1 - P_{\sqrt{M}})^2$$

where $P_{\sqrt{M}}$ is the symbol error probability of a $\sqrt{M}$ -ary AM with one-half the average power of the QAM signal. From (8.13) we have

$$P_{\sqrt{M}} = \frac{2(\sqrt{M} - 1)}{\sqrt{M}} Q \left(\sqrt{\frac{3E_{avg}}{(M - 1)N_o}} \right) \quad (8.53)$$

where E_{avg}/N_o is the average SNR per symbol. The symbol error probability of the

square QAM is

$$P_s = 1 - (1 - P_{\sqrt{M}})^2 = 2P_{\sqrt{M}} - P_{\sqrt{M}}^2 \quad (8.54)$$

At high SNR,

$$P_s \cong 2P_{\sqrt{M}} = \frac{4(\sqrt{M} - 1)}{\sqrt{M}} Q \left(\sqrt{\frac{3E_{avg}}{(M-1)N_o}} \right) \quad (8.55)$$

Note that (8.54) is exact for square QAM with $M = 2^k$ where k is even. When k is odd there is no equivalent $\sqrt{M}$ -ary AM system. However, we can find a tight upper bound [7, Page 655]

$$\begin{aligned} P_s &\leq 1 - \left[1 - 2Q \left(\sqrt{\frac{3E_{avg}}{(M-1)N_o}} \right) \right]^2 \\ &\leq 4Q \left(\sqrt{\frac{3kE_{bavg}}{(M-1)N_o}} \right) \end{aligned} \quad (8.56)$$

for any $k \geq 1$, where E_{bavg}/N_o is the average SNR per bit.

To obtain bit error probability from the symbol error probability, we observe that square QAM can be perfectly Gray coded. That is, there is only one bit difference between adjacent symbols. Each symbol error most likely causes one bit error at large SNR. Thus

$$P_b \cong \frac{P_s}{\log_2 M} \quad (8.57)$$

Figure 8.13 shows the P_b curves for $M = 4, 8, 16, 32, 64, 128$, and 256 where the curves for $M = 8, 32$, and 128 are tight upper bounds (dotted lines).

In the following we compare QAM with MPSK. From the P_s of MPSK (4.24) and (8.55) or (8.56), the ratio (QAM over MPSK) of the arguments inside the square root sign of the Q-function is

$$R_M = \frac{3}{2(M-1) \sin^2 \pi/M} \quad (8.58)$$

This reflects the ratio of the signal power. This ratio is tabulated in Table 8.2.

From the table we can see that for $M > 4$, the QAM is superior to MPSK. Furthermore, for large $M (\geq 32)$ the power savings increases 3 dB for doubling the number of signal points. This can be explained from (8.58) by observing that $R_M \cong 3M/2\pi^2$ for large M . This can also be looked at from another point of view. Examining (8.55) or (8.56) reveals that for large M , doubling M incurs 3 dB penalty

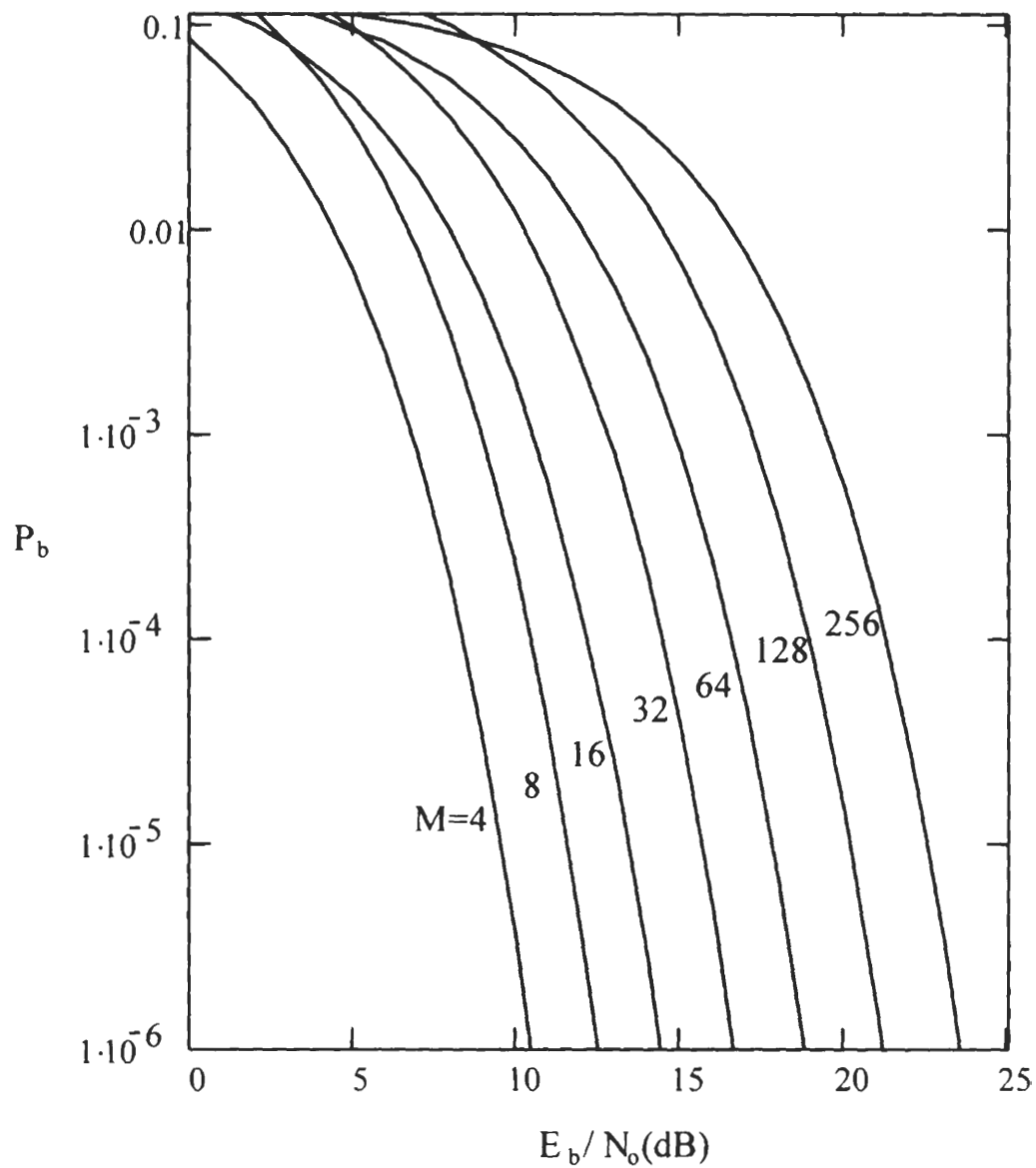


Figure 8.13 Square QAM bit error probability.

M	R_M	$10 \log R_M$ (dB)
4	1	0
8	1.463	1.65
16	2.627	4.20
32	5.036	7.02
64	9.889	9.95
128	19.61	12.92
256	39.06	15.92

Table 8.2 Power savings of QAM over MPSK.

in signal power, whereas the penalty for MPSK and MAM is 6 dB as we pointed out before. That is why Table 8.2 shows a 3 dB increase in power savings for doubling M when QAM is compared with MPSK (and it is also true compared with MAM).

8.8 SYNCHRONIZATION

Clock synchronization for QAM is usually achieved by clock recovery circuit which extracts the clock from the demodulated signal or uses the demodulated signal to control the local oscillator. Clock recovery techniques for QAM are the same as those for MPSK. The open-loop clock recovery circuits in Figure 4.37 and the early/late-gate clock recovery circuit in Figure 4.38 are all applicable to QAM. Refer to Chapter 4 for details.

Carrier synchronization is always necessary for square QAM constellations, even when differential coding is used. This is because differential coding for the square QAM is only for some of the bits in a symbol. Even though these bits can be determined by comparing two consecutive symbols (differential demodulation), the rest of the bits still must be determined by coherent demodulation. Thus the entire symbol might as well be coherently demodulated. For a description of differential coding for QAM, refer to the next section. Circular QAM constellations do not require carrier synchronization if differential encoding is used. We will discuss this in Chapter 10.

Carrier synchronization can be achieved by the pilot-tone technique or a separate synchronization channel, which requires extra bandwidth, as we mentioned in Chapter 4. Again we will not elaborate on this. Instead we focus on the carrier recovery techniques. There are two major types of carrier recovery techniques for QAM. One is the fourth-power loop (or times-four loop) and another is the decision-directed carrier recovery (DDCR). According to [8, Page 182], the decision-directed carrier recovery technique is one of the most popular carrier recovery schemes used in fixed-link

QAM systems. The fourth-power loop carrier recovery has been suggested for digital radio in fading environment [9]. In the following we first discuss the fourth-power loop technique and then the decision-directed carrier recovery technique.

Recall that the M th-power loop (Figure 4.35) is used for MPSK, but for M -ary QAM, the M th-power loop is not necessary, only the fourth-power loop is needed. Unfortunately the squaring loop does not work for symmetrical QAM since the average energy at $2f_c$ is zero. It is also known that for M greater than four, the data pattern effects lead to an increased amount of carrier phase jitter in an M th-power loop [9]. Thus the fourth-power loop is the right choice for QAM.

We show below that the squaring loop does not work for symmetrical QAM, but the fourth-power loop does. From Section 8.5 we see that A_{i1} and A_{i2} are generated by level generators using the input data which are random. Therefore A_{i1} and A_{i2} are random. We assume that their values are symmetrical about zero, that is, they have zero means. We further assume that the input data are ergodic random processes. Consequently A_{i1} and A_{i2} are ergodic random processes so that the time average is equal to the statistical average. All the above assumptions are the usual case in digital communications. We square the QAM signal given in (8.23) and take the average over all possible i to obtain

$$\begin{aligned} E\{s_i^2(t)\} &= E\{p^2(t)[A_{i1}^2 \cos^2 \omega_c t - 2A_{i1}A_{i2} \cos \omega_c t \sin \omega_c t + A_{i2}^2 \sin^2 \omega_c t]\} \\ &= E\{p^2(t)[A_{i1}^2 \cos^2 \omega_c t + A_{i2}^2 \sin^2 \omega_c t]\} \\ &= \frac{1}{2}p^2(t) [(E\{A_{i1}^2\} + E\{A_{i2}^2\}) + (E\{A_{i1}^2\} - E\{A_{i2}^2\}) \cos 2\omega_c t] \end{aligned}$$

where $\omega_c = 2\pi f_c$ and the middle term in the first expression vanishes since A_{i1} and A_{i2} are independent zero-mean random processes so that

$$E\{A_{i1}A_{i2}\} = E\{A_{i1}\}E\{A_{i2}\} = 0$$

For symmetrical QAM, $E\{A_{i1}^2\} = E\{A_{i2}^2\}$, the above becomes

$$E\{s_i^2(t)\} = p^2(t)E\{A_{i1}^2\}$$

which has no periodical component at f_c or its multiples. We can intentionally unbalance the QAM constellation to make carrier recovery possible, but this leads to inefficient signal constellations. A fourth-power nonlinearity can produce a nonzero component at $4f_c$ even if the QAM is symmetrical. Using trigonometrical identities we can show that (see Appendix 8A at the end of this chapter)

$$\begin{aligned} E\{s_i^4(t)\} &= \frac{1}{8}p^4(t)[E\{A_{i1}^4\}(3 + 4\cos 2\omega_c t + \cos 4\omega_c t) \\ &\quad + E\{A_{i2}^4\}(3 - 4\cos 2\omega_c t + \cos 4\omega_c t)] \end{aligned}$$

$$+\frac{3}{4}p^4(t)E\{A_{i1}^2\}E\{A_{i2}^2\}(1-\cos 4\omega_c t) \quad (8.59)$$

For symmetrical QAM, since $E\{A_{i1}^2\} = E\{A_{i2}^2\}$, we have

$$\begin{aligned} E\{s_i^4(t)\} &= \frac{3}{4}p^4(t) [E\{A_{i1}^4\} + (E\{A_{i1}^2\})^2] \\ &\quad + \frac{1}{4}p^4(t) [E\{A_{i1}^4\} - 3(E\{A_{i1}^2\})^2] \cos 4\omega_c t \end{aligned} \quad (8.60)$$

There is a nonzero component at $4f_c$.

For the square QAM, $A_{i1} = I_i \sqrt{E_0/E_p}$, $A_{i2} = Q_i \sqrt{E_0/E_p}$ (see (8.41)).

$$E\{A_{i1}^2\} = \frac{E_0}{E_p} \frac{1}{3}(M-1) \quad (8.61)$$

where the result of (8.44) is used, and

$$E\{A_{i1}^4\} = \left(\frac{E_0}{E_p}\right)^2 E\{I_i^4\}$$

where

$$\begin{aligned} E\{I_i^4\} &= \frac{1}{L} [(-(L-1))^4 + (-(L-3))^4 + \cdots + (L-3)^4 + (L-1)^4] \\ &= \frac{2}{L} [1^4 + 3^4 + \cdots + (L-1)^4] \\ &= \frac{1}{15} (3L^4 - 10L^2 + 7) = \frac{1}{15} (3M^2 - 10M + 7) \end{aligned} \quad (8.62)$$

Substitute (8.61) and (8.62) into (8.60), we have

$$E\{s_i^4(t)\} = \frac{1}{30}p^4(t) \left(\frac{E_0}{E_p}\right)^2 [(7M^2 - 20M + 13) - (M^2 - 1) \cos 4\omega_c t] \quad (8.63)$$

From (8.63) we can see that the $4f_c$ component always has a phase of π for the square QAM.

For the square 16-QAM we have

$$E\{s_i^4(t)\} = \frac{1}{2}p^4(t) \left(\frac{E_0}{E_p}\right)^2 (99 - 17 \cos 4\omega_c t)$$

M	r_M	$20 \log R_M$ (dB)
4	0.333	-9.54
16	0.172	-15.30
64	0.149	-16.51
256	0.144	-16.80

Table 8.3 Amplitude ratio.

If $p(t) = 1$, then $E_p = T$, then

$$E\{s_i^4(t)\} = \frac{1}{2} \left(\frac{E_0}{T} \right)^2 (99 - 17 \cos 4\omega_c t)$$

This shows that the $4f_c$ component has a phase of π and an amplitude of $17/99 \cong 17.2\%$ of the dc component. In fact we can calculate this ratio for a series of M , as listed in Table 8.3.

$$r_M = \frac{M^2 - 1}{7M^2 - 20M + 13} \quad (8.64)$$

Note that the above results (Table 8.3) are for ideal fourth-power devices and square QAM. For other types of constellations and nonideal devices, the results would be different. For some types of constellations, the $4f_c$ component may be too weak to be useful. Then other carrier recovery techniques, such as the decision-directed carrier recovery technique may be needed.

The block diagram in Figure 4.35 can be used for the fourth-power loop by letting $M = 4$. A slightly different version was proposed by Rustako et al [9] where the fourth-power loop was proposed for QAM digital radio receivers. Figure 8.14 is the diagram given in [9]. The VCO is working at f_c . A $\times 4$ block is inserted in the PLL loop to raise the VCO frequency to $4f_c$, which is then compared with the output of another $\times 4$ block whose input is the received signal. The $\times 4$ block is actually realized by a frequency-doubler whose nonlinearity generates a $4f_c$ component as a by-product at a level of about 20 dB below the $2f_c$ component. A detailed analysis of the times-four loop can be found in [9]. Rustako notes that in a fading environment, decision-directed carrier recovery suffers when occasional receiver outage destroys the accuracy of the data detections. This causes loss of carrier recovery and considerable time may be required to re-acquire the carrier. He notes that times-four carrier recovery does not depend on data decision. For this and other reasons, times-four carrier recovery was proposed for digital radio receivers.

As mentioned, the decision-directed carrier recovery technique is one of the most popular carrier recovery schemes used in fixed-link QAM systems. The mech-

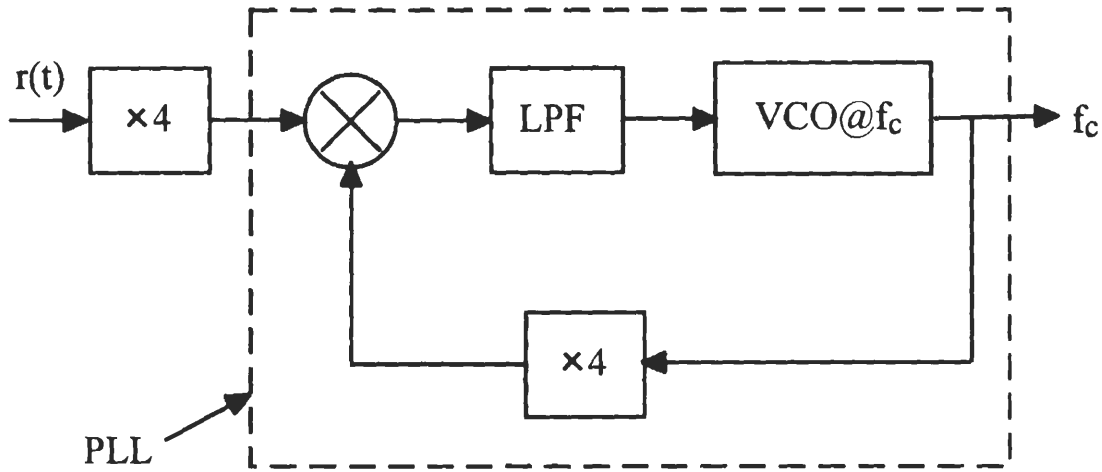


Figure 8.14 Times-four carrier recovery loop for QAM.

anism of DDCR is not based on nonlinearity. In this scheme, when a symbol is received, it is demodulated and a decision is made as to which was the most likely constellation point transmitted. Then it is assumed that the phase difference between the received symbol and the constellation point is due to carrier-recovery error, and the carrier recovery is updated accordingly. DDCR has the advantage that it can be used for all types of constellations. However, it exhibits a BER threshold. If the receiver BER is less than this threshold, it works extremely well since the “decisions” that assist the carrier recovery are correct almost all the time. If the receiver BER is higher than this threshold, the decisions on the constellation points are wrong more frequently. The update signal to the carrier recovery system therefore is erroneous more frequently, the cumulative effect can drive the carrier phase further and further away from the correct value. Whether DDCR is suitable for a particular application depends on this BER threshold.

Horikawa et al designed a DDCR system for a 200Mbps square 16-QAM [10]. In their design, the square 16-QAM constellation points are divided into two subgroups, called class I and class II phasors (Figure 8.15). The class II phasors are those on the second circle with a normalized amplitude of $\sqrt{10}$. The rest are the class I phasors, four are on the outer circle, with an amplitude of $\sqrt{18}$; four are on the inner circle, with an amplitude of $\sqrt{2}$. The quadrature components of the class I phasors are equal in magnitude (i.e., $|I| = |Q|$), whereas the quadrature components of the class II phasors have a 3 to 1 ratio (i.e., $|I| = 3|Q|$ or $|Q| = 3|I|$).

Horikawa’s DDCR system operates only on class I phasors. In other words, only class I phasors in the received signal are utilized by the DDCR system to recover the carrier. The DDCR system compares the values of the demodulated signal I and Q.

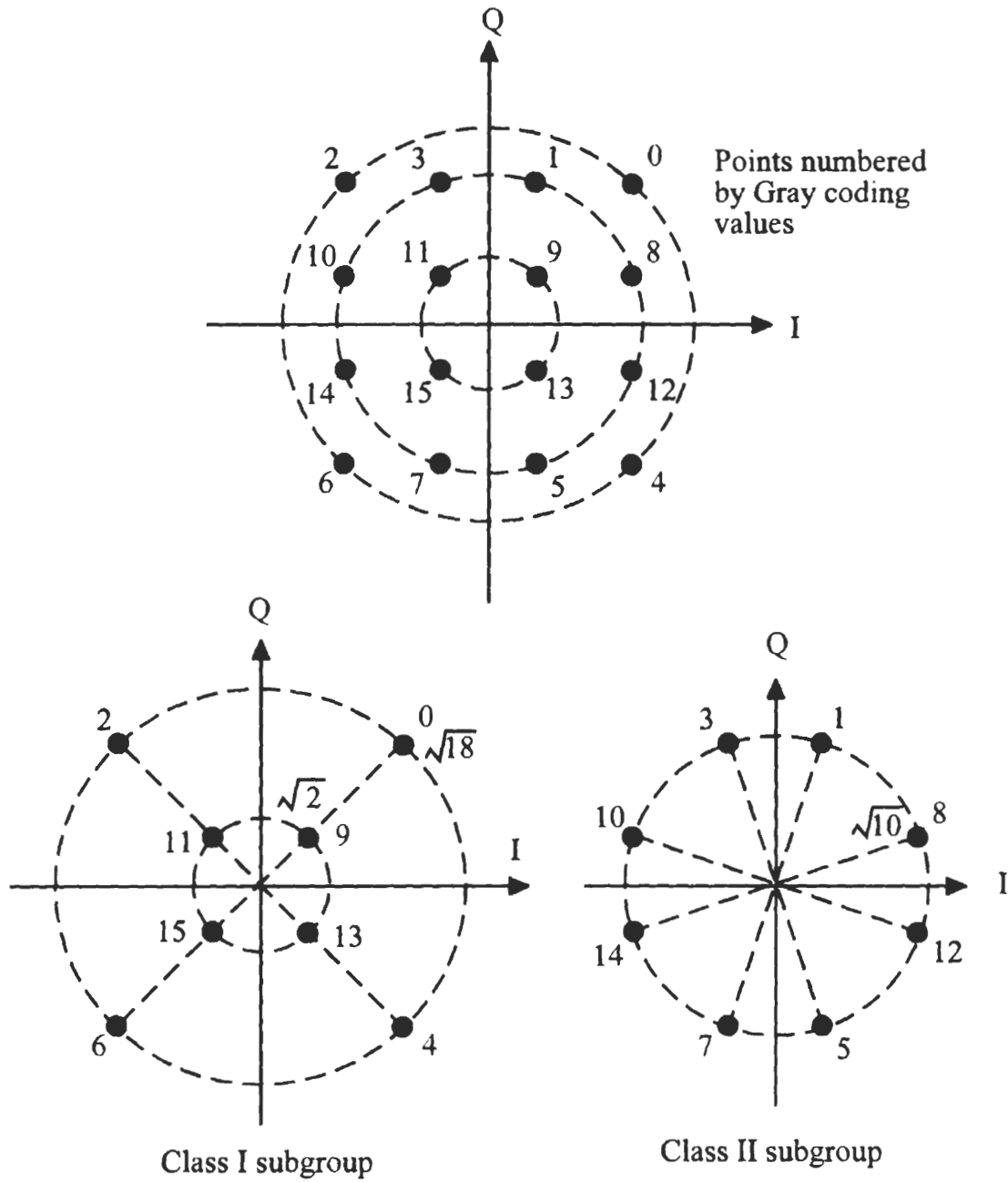


Figure 8.15 Square 16-QAM subgroups.

If they satisfy the relationship

$$\frac{|I|}{2} < |Q| < 2|I| \quad (8.65)$$

then the system decides that a class I phasor has been sent. This rule is designed to effectively reject class II phasors while detecting class II phasors in a noisy environment. Once a class I phasor is detected, its I and Q values are used to determine polarity of the carrier drift and subsequently to adjust the VCO. The algorithm is as follows [8]. First compute the quantities

$$\begin{aligned} a &= \text{pol}(I) \oplus \text{pol}(Q) \\ b &= \text{pol}(I + Q) \\ c &= \text{pol}(I - Q) \\ d &= (a \oplus b) \oplus c \end{aligned} \quad (8.66)$$

where $\oplus$ denotes modulo-2 addition or XOR operation, and

$$\text{pol}(x) = \begin{cases} 1, & x > 0 \\ 0, & x < 0 \end{cases}$$

The d is used to control the VCO.

According to the above algorithm, the values of (abc) and d are shown in Figure 8.16 for each octant. As can be seen, the value of d changes between 0 and 1 alternately from an octant to the next. The quantity d is used to adjust the VCO frequency and phase. The demodulated signal is

$$r = \text{LPF}\{A_i \cos(\omega_c t + \theta_i) \cdot 2 \cos(\omega_c t + \Delta\theta)\} = A_i \cos(\theta_i - \Delta\theta) \quad (8.67)$$

where LPF denotes low-pass filtering. The VCO is assumed to have an amplitude of 2 and a phase error $\Delta\theta$ which is due to frequency error and initial phase error. From Figure 8.16, when $d = 0$, the angle of r is less than the transmitted phasor. From (8.67), this implies that the VCO has a positive phase error. Thus the VCO is instructed to run slower (reduce its frequency). When $d = 1$, the angle of r is greater than the transmitted phasor. From (8.67), this implies that the VCO has a negative phase error. Thus the VCO is instructed to run faster (increase its frequency). For example, assume $r = (I, Q) = (2.24, 3.6)$ is received (see Figure 8.16). Then $a = \text{pol}(2.24) \oplus \text{pol}(3.6) = 1 \oplus 1 = 0$, $b = \text{pol}(5.84) = 1$, $c = \text{pol}(-1.36) = 0$, and $d = 0 \oplus 1 \oplus 0 = 1$. Thus the VCO is instructed to increase its frequency.

The DDCR system is shown in Figure 8.17. The received signal $r(t)$ is demodulated by the I - and Q -branches to produce I and Q . Then I and Q are fed into four comparators to generate a , b , and c , which are fed into XORs to generate d . At the

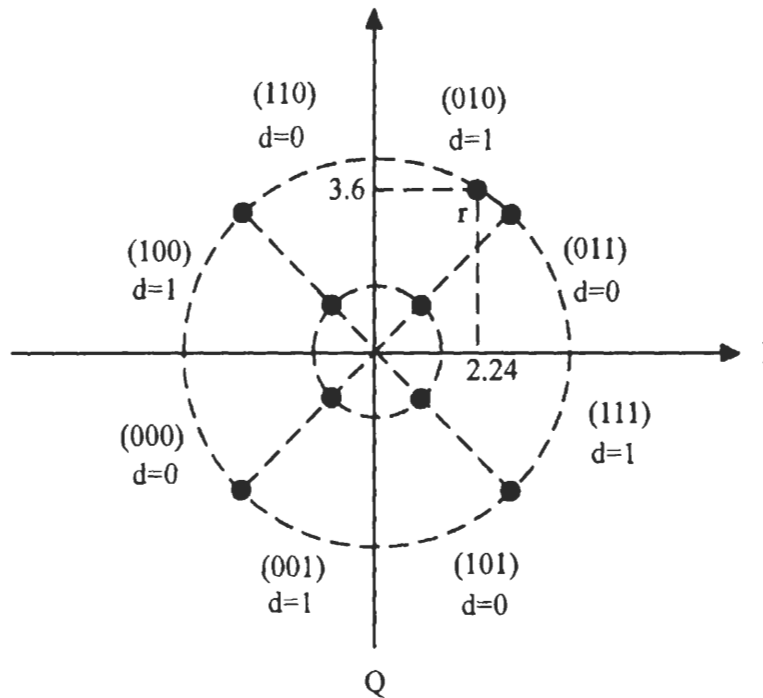


Figure 8.16 Distribution of the quantities in the DDCR algorithm.

same time, the I and Q are full-wave rectified to produce $|I|$ and $|Q|$, which are passed through the other two comparators to check whether they satisfy the relation (8.65). If they do, the selective gate will open, allowing signal d to pass the D flip-flop to control the VCO. Once synchronization is established in Figure 8.17, the I - and Q -channel outputs can be tapped by two threshold detectors to determine the transmitted symbol. Thus only two extra threshold detectors are needed to make Figure 8.17 a complete demodulator.

8.9 DIFFERENTIAL CODING IN QAM

In Chapter 4 we studied differential coding for MPSK signals. Similarly, differential coding is needed for QAM. It is needed for square QAM to resolve the phase ambiguity in carrier recovery. For circular QAM the use of differential coding can eliminate the need for carrier recovery, which is particularly attractive for fading channels. We will elaborate on this in Chapter 10. For now we focus on how differential coding is constructed in square QAM.

Weber proposed a scheme for differentially encoding the signals for QAM and

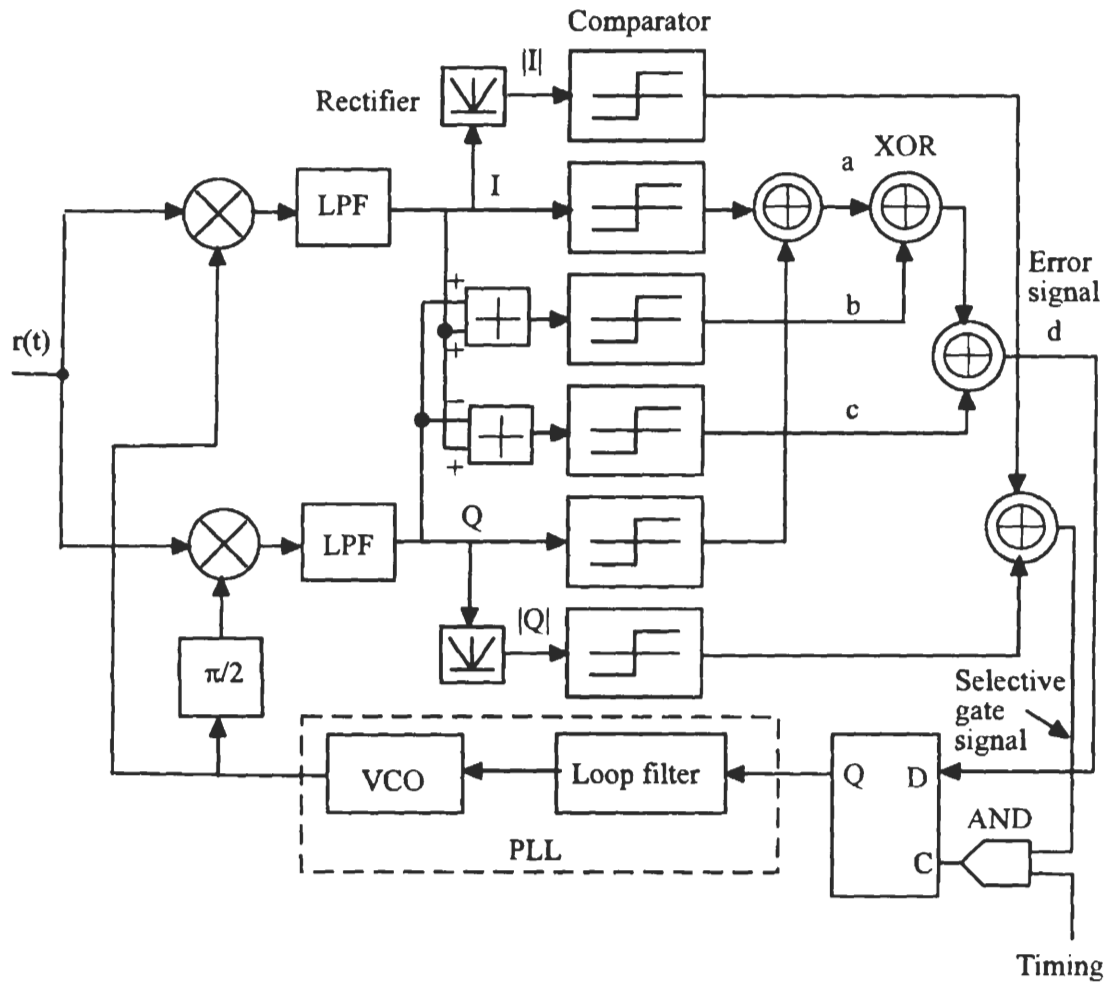


Figure 8.17 Implementation of a decision-directed carrier recovery system for square 16-QAM. From [10]. Copyright © 1979 IEEE.

MPSK [11]. We follow his paper here. First, a signal set has an L -fold rotational symmetry if the signal set pattern remains unchanged after a rotation of $\pm I \cdot (2\pi/L)$ radians, where I and L are integers. For example, in Figure 8.5, the type I constellation has an 8-fold symmetry, the type II constellation has a 4-fold symmetry, and the type III constellation has also a 4-fold symmetry. As a matter of fact, all square QAM schemes are 4-fold symmetrical. It is also clear that all MPSK schemes are M -fold symmetrical. The L -fold rotational symmetry causes a phase ambiguity of $2\pi/L$, that is, any phase rotation of $2\pi/L$ or its multiples in the received signal will cause the signal to be demodulated as another signal. For example, for the square QAM, times-four loop is used for carrier recovery. Any phase rotation of $\pi/2$ in the received signal will not change the constellation, thus the recovered carrier, which is based on the average of signals over the constellation (over a long period of time in practice), will always have the same phase. Thus the signals in the first quadrant may be demodulated as the signals in the second quadrant, or third quadrant, or fourth quadrant, depending on the actual amount of phase rotation. Differential coding is an efficient way to resolve the phase ambiguity. Other methods include sending a separate synchronizing signal or periodic insertion of a synchronizing sequence in the data sequence. The differential coding method does not require extra bandwidth like the other two, but increases the error probability slightly.

As discussed before, the QAM signal points are usually Gray coded in order to minimize bit error probability. Differential coding will violate the Gray code rule, thus increasing the bit error probability. However, the differential coding procedure should be carefully chosen such that it can remove the symmetrical ambiguities at a minimum increase of the error probability. The following is the differential coding procedure given by Weber for an $M = 2^K$ signal set for which $L = 2^N$.

1. Divide the signal space into L equal pie-shaped sectors. These sectors are differentially encoded using the first N bits of each K bits of data. That is, the first N bits determine the *change in sector*.
2. The remaining $K - N$ bits determine the signal point within the sector. The $2^{(K-N)}$ signal points within the sector must be Gray coded to reduce the probability of error. (However, perfect Gray coding may not be possible.)

We will see how this is done through an example. Figure 8.18 is the differential encoding for the square 16-QAM. In the figure, the signal space is divided by the I- and Q-axes into $L = 4$ pie-shaped sectors (quadrants in this case). Signal points are labeled with the last two bits in a 4-tuple. As far as these two bits are concerned, they are Gray coded in each sector. In addition they are arranged so that they are 4-fold symmetrical in terms of the last two bits. That is, a rotation of $\pi/2$ or its multiples of any signal will result in a signal having the same last two bits. The first two bits are used to determine the change of quadrant. Table 8.4 shows the encoding

Encoding	ref.				
Message	0000	1011	0001	1100	0110
Δq		-1	0	+2	+1
q_i	q_1	q_4	q_4	q_2	q_3
Transmitted signal	s_1	s_{16}	s_{14}	s_5	s_{11}
Decoding ($\Delta\phi = 0$)	ref.				
Received signal	s_1	s_{16}	s_{14}	s_5	s_{11}
Δq		-1	0	+2	+1
Decoded Message	0000	1011	0001	1100	0110
Decoding ($\Delta\phi = \pi/2$)					
Received signal	s_5	s_4	s_2	s_9	s_{15}
Δq		-1	0	+2	+1
Decoded Message	0000	1011	0001	1100	0110

Table 8.4 Examples of differential coding for square 16-QAM.

the same time the reference signal is also changed to s_5 due to the rotation. Signal s_4 is compared with the previous signal (s_5) to determine the quadrant change (-1). This determines 10 as the first two bits. The last two bits then are determined as 11 by the fact that s_4 is labeled 11. From the table we can see that phase rotation does not affect decoding of the signals. Thus the phase ambiguity is removed.

For general K -bit (K even) square QAM, again only the first two bits need to be differentially encoded for resolving the quadrant ambiguity, while the remaining $K - 2$ bits are Gray coded in each quadrant. For other constellations, such as circular QAM, the same encoding principles apply. The concrete encoding rules depend on the concrete constellation structure. See [11] for more examples.

In evaluating the error probability of the differentially coded QAM, a high SNR is assumed. This implies the errors are dominantly the errors made between points separated by the minimum distance $d_{\min}$. Thus we only consider this type of error in approximating the error probability. The above differential encoding schemes produce two types of symbol errors: those between symbols within the same sector and those between symbols of different sectors. In the first type of error, the first N bits are not affected since they are the same for all points in the sector. Thus only one or more of the $(K - N)$ bits will be in error, depending on how the symbols in the sector have been Gray coded. If the symbols in the sector are perfectly Gray coded, like in the square 16-QAM case, only one bit will be in error. If the symbols in the sector are not perfectly Gray coded, more than one bit on average will be in error. Then we say there is a Gray code penalty for the imperfect Gray coding. For this type of error, no error propagation exists since the differentially encoded bits are unaffected. In the second type of error, bit errors can arise from two sources. First, since the sec-

tor boundary has been crossed, a minimum of two bit errors will occur in the first N bits, one in each of two succeeding symbols (error propagation), due to the comparisons used in differential decoding. Additional bit errors may occur in the remaining $K - N$ bits depending on how these adjacent border symbols are encoded. While a perfect Gray code may be found for the symbols within a sector, a Gray code penalty may exist with symbols lying on the sector boundaries.

The relationship between the error probability of the differentially encoded system and that of the uncoded system can be given in the following general formula

$$P_{b,d} = F \cdot P_b$$

where $P_{b,d}$ is the bit error probability of the differentially coded system, and P_b is the bit error probability of the differentially uncoded system. The factor F is the penalty for differential encoding. For the differentially uncoded system,

$$P_b = \frac{g}{K} P_s$$

where P_s is the symbol error probability, $g \geq 1$ is the Gray code penalty, representing the average number of erroneous bits in a symbol error. For the perfect Gray coded system, $g = 1$. For the differentially encoded system,

$$P_{b,d} = \frac{f}{K} P_s$$

where f is the differentially encoded Gray code penalty, also representing the average number of erroneous bits in a symbol error. Thus the penalty for differential encoding is

$$F = \frac{f}{g} \quad (8.68)$$

The factor f can be found as follows.

1. Within a single sector draw lines between symbol points pairs separated by $d_{\min}$; denote the total number of these lines by N_1 ;
2. Next to each line write the Hamming distance (number of bits which differ) between the two signal points; denote the sum of all such Hamming distances in the sector by H_1 ;
3. Draw lines between points lying on *one* of the sector boundaries with points in the adjacent sectors separated by $d_{\min}$; denote the number of these lines by N_2 and denote the sum of their Hamming distances by H_2 .

Then the differential encoding penalty is

$$f = \frac{H_1 + H_2 + 2N_2}{N_1 + N_2} \quad (8.69)$$

where the third term in the numerator is due to error propagation in the differentially encoded bits. The numerator is in fact the total number of bit errors when all possible symbol errors are considered, and the denominator is in fact the total number of possible symbol errors. Thus f is the average bit errors per symbol. Note that H_1/N_1 is the Gray code penalty within a sector and H_2/N_2 is the Gray code penalty across the boundary.

For the differentially encoded square 16-QAM in Figure 8.18, $N_1 = 4$ and $H_1 = 4$ while $N_2 = 2$ and $H_2 = 2$. Thus $f = (4 + 2 + 4)/(4 + 2) = 10/6 = 1.67$. For general K -bit (K even) square QAM with the first two bits differentially encoded for resolving the quadrant ambiguity and the remaining $K - 2$ bits Gray coded, it can be shown that

$$\begin{aligned} N_1 &= 2^{K-1} - 2^{K/2} \\ H_1 &= 2^{K-1} - 2^{K/2} \\ N_2 &= 2^{(K/2-1)} \\ H_2 &= \left(\frac{K}{2} - 1 \right) 2^{(K/2-1)} \end{aligned} \quad (8.70)$$

For the above general case $g = 1$ so that from (8.70) and (8.69)

$$F = 1 + \frac{K/2}{2^{K/2} - 1}, \quad (\text{square QAM, } K \text{ even})$$

Thus F goes from two for $K = 2$ (4-QAM or QPSK) to nearly one for very large K . This penalty of two or less is insignificant in term of increase in SNR, which is usually a fraction of a dB.

8.10 SUMMARY

We discussed M-ary amplitude modulation (MAM) in Section 8.1. In a concise fashion, we covered all aspects of MAM, including PSD, optimum detection, error probability, modulator and demodulator. The discussion was general in that the results are applicable to both baseband and bandpass cases, and in the case of bandpass MAM, pulse shaping was also included. We also showed the equivalence between the correlator receiver and matched filter receiver even when pulse shaping is involved. There was a subsection devoted to OOK. It is often the first modulation scheme introduced